Software Testing

ISTQB / ISEB Foundation Exam Practice

1 Principles	2 Lifecycle	3 Static testing
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Test Techniques

Chapter 4

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- Categories of Test Techniques
- Black-box Test Techniques
- White-box Test Techniques
- Experience-based Test Techniques

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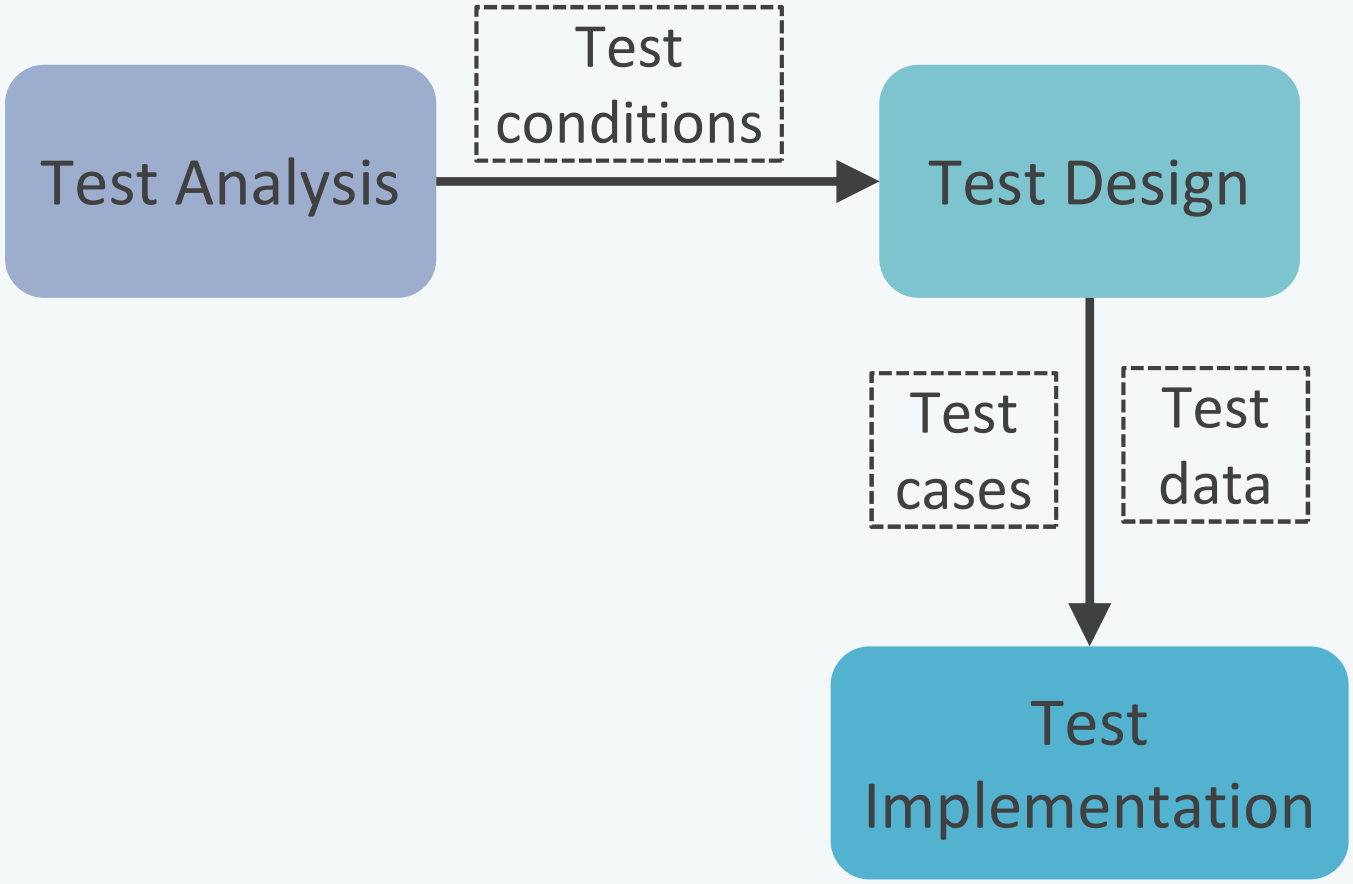
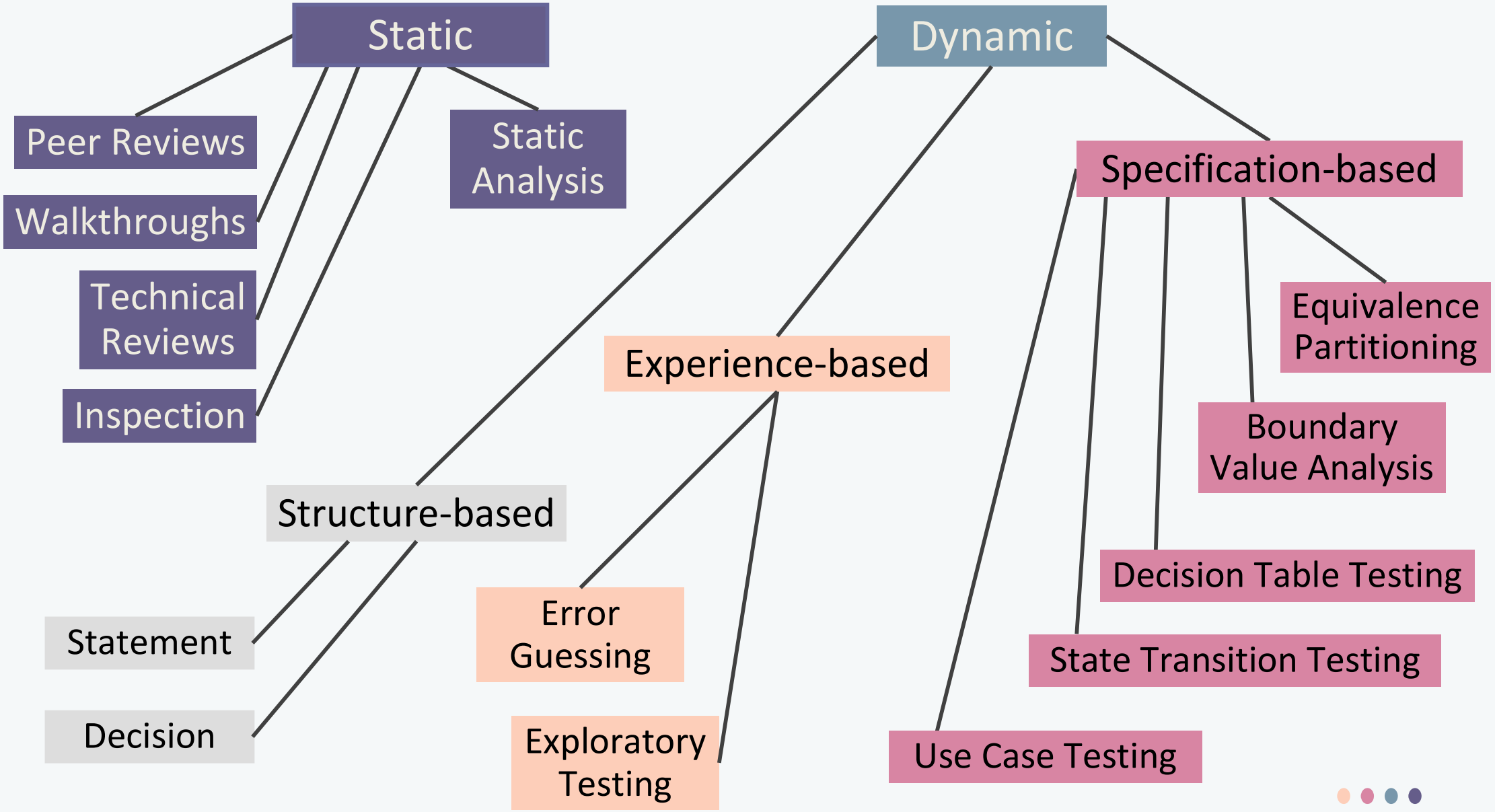
- Categories of Test Techniques
- Black-box Test Techniques
- White-box Test Techniques
- Experience-based Test Techniques

Categories of Test Techniques

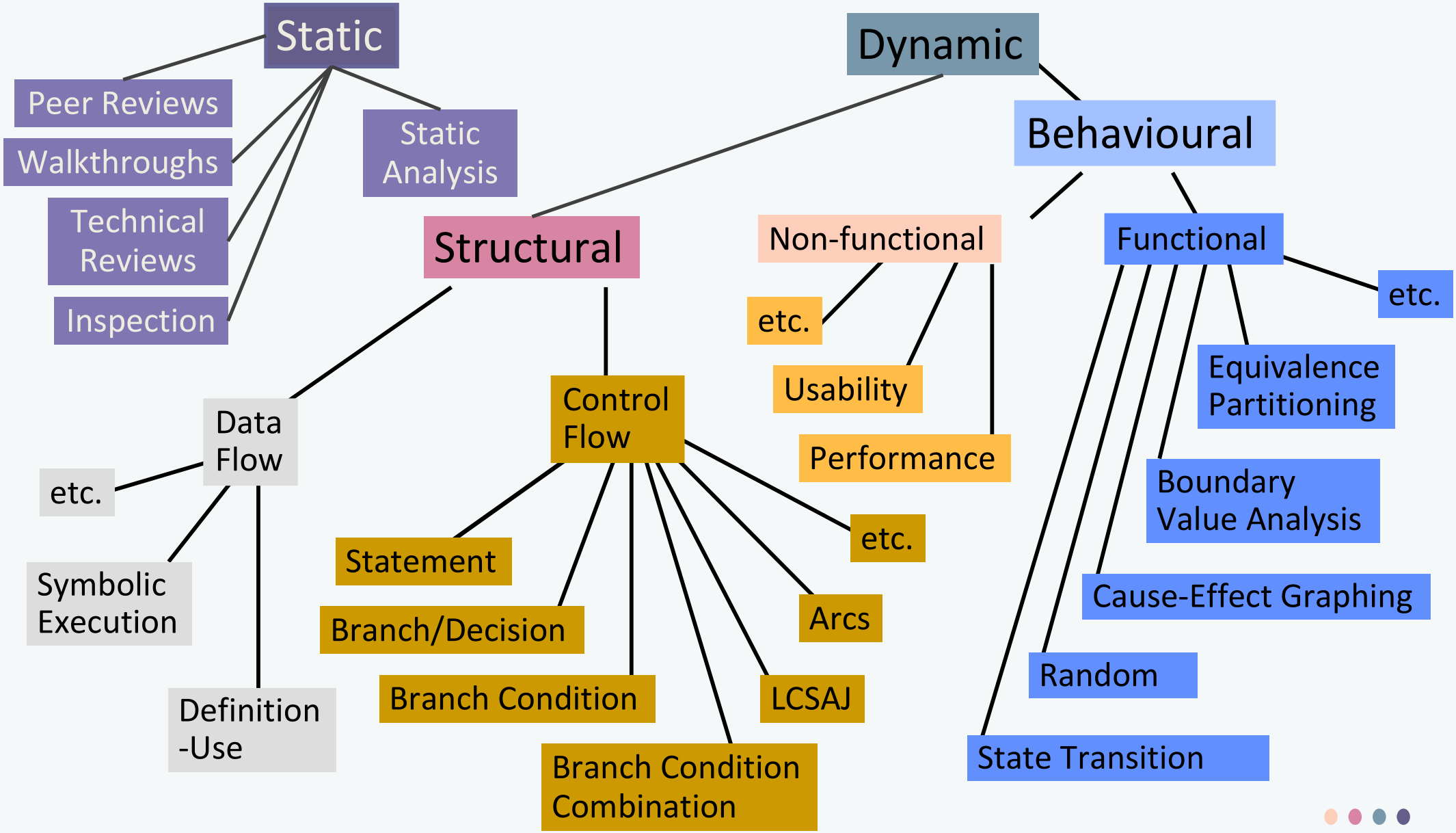
- The purpose of a test technique is to identify test conditions, test cases, and test data.



Categories of Test Techniques



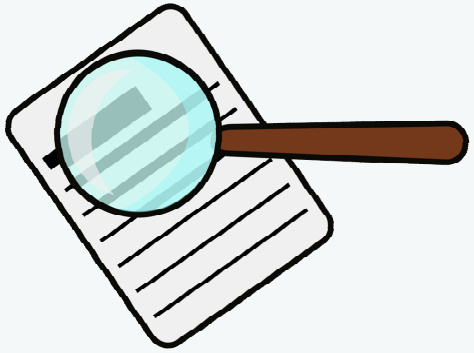
Categories of Test Techniques



Three Types of Systematic Test Techniques

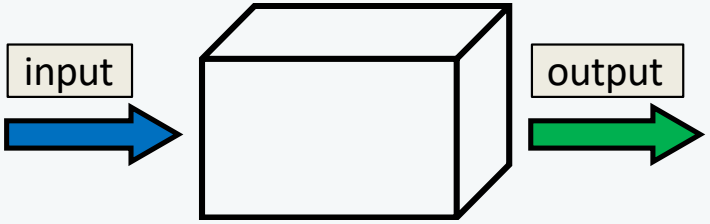
Static (non-execution)

- examination of documentation, source code listings, etc.



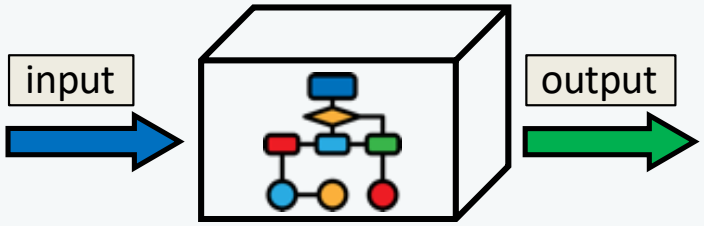
Behavioural (Black Box)

- based on behaviour / functionality of software



Structural (White Box)

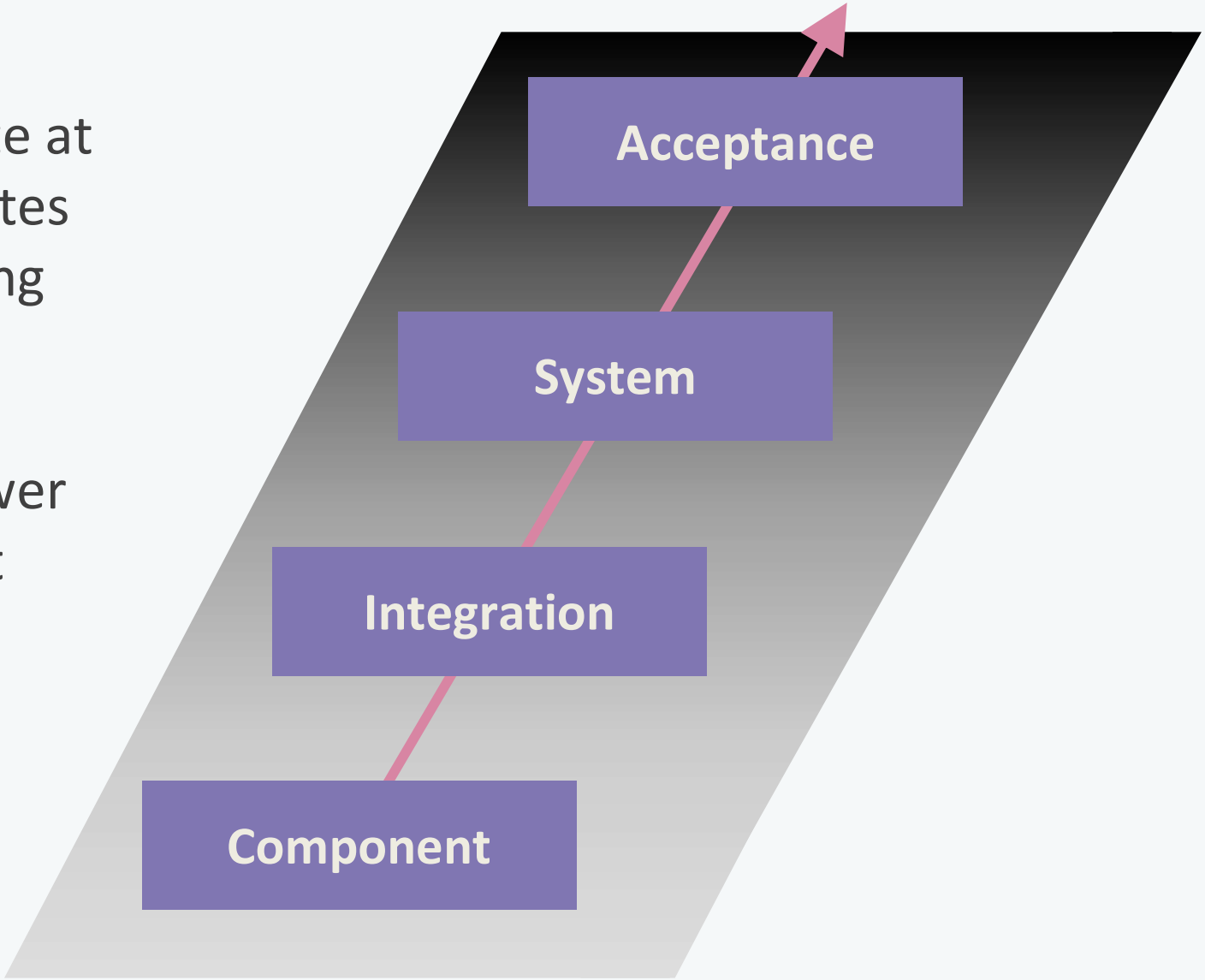
- based on structure of software



Black box vs. White box

Black-box appropriate at all levels but dominates higher levels of testing

White-box used predominately at lower levels to compliment black box



Dynamic Testing: 3 Sub-categories

	Black-box	White-box	Experience-based
Test basis	SW reqs. Specifications. Use cases. User stories.	Architecture analysis. Detailed design. Internal structure. Code.	Knowledge of (Testers, Developers, Users).
Utilisation	Detect gaps b/w reqs. & implementation of reqs, deviations from reqs.	Determine paths through the software that either were taken or need to be taken.	Complement the other two types. Used when there is no / little / outdated specifications.
Coverage	Based on items tested on the test basis & technique applied to the test basis.	Based on items tested within a selected structure.	N/A

Question

Which one of the following options is categorized as a black-box test technique?

A. Techniques based on analysis of the architecture

B. Techniques checking that the test object is working according to the technical design

C. Techniques based on the expected use of the software

D. Techniques based on formal requirements

Question

If test cases are derived from looking at the code, what type of test design technique is being used?

A. Black-box

B. White-box

C. Specification-based

D. Behaviour-based

Question

Which of the following test techniques uses the requirements specifications as a test basis?

A. Structure-based

B. Black-box

C. White-box

D. Exploratory

CONTENTS

- Categories of Test Techniques
- Black-box Test Techniques
  - Equivalence Partitioning
  - Boundary value analysis
  - Decision table testing
  - State transition testing
- White-box Test Techniques
- Experience-based Test Techniques

Equivalence Partitioning (EP)

- Can be applied at any level of testing & often a good technique to use first
- Idea: divide (i.e., to partition) a set of test conditions into groups/sets where all elements of the set can be considered the same – the system should handle them equivalently.
- Assumption: if one value works, all will work.
- One from each partition better than all from one

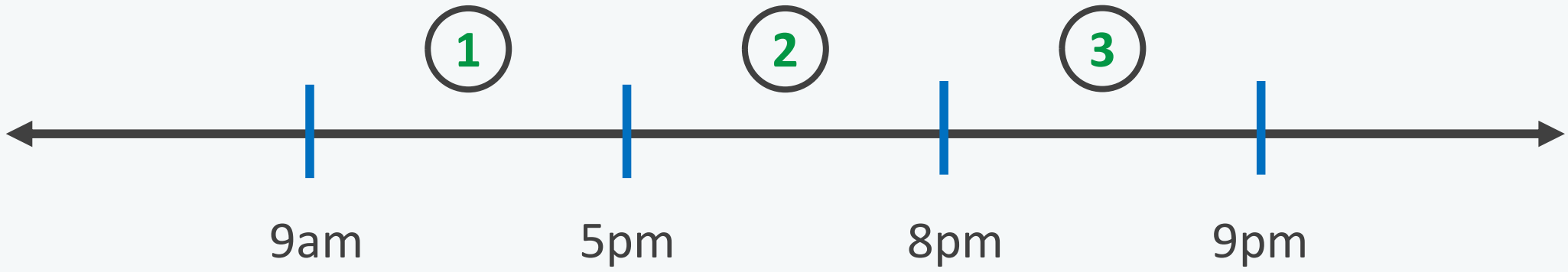
Equivalence Partitioning (EP)

Example. In the SRS, we have FR08 about registration of a special training room in a training centre (opening 9am – 10pm). An SE course is currently always run from 5pm – 7:59pm everyday in that room. We are testing the booking function of that room.

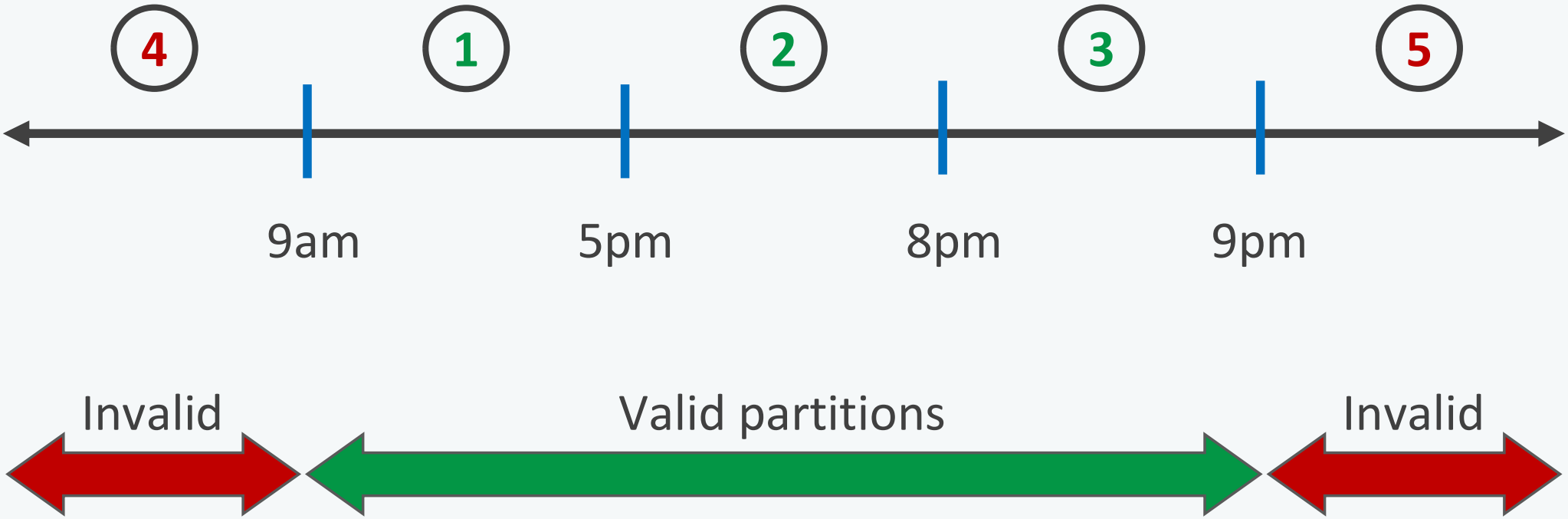
If someone tries to book the room from 9am – 5pm, the room is available

If he/she tries to book the room from 5pm – 7:59pm, the room is occupied

If he/she tries to book the room from 9pm – 10pm, the room is available



Equivalence Partitioning (EP)



Equivalence Partitioning (EP)

Example: A savings account in a bank earns a different rate of interest depending on the balance in the account. To test the function that calculates the interest due, we can have different balance ranges:

[$0 … $100] – 3% interest rate

($100 … $1000) – 5% interest rate

[$1000 …] – 7% interest rate

Valid (for 3% interest)		Valid (for 5%)		Valid (for 7%)
$0.00	$100.00	$100.01	$999.99	$1,000.00

Equivalence Partitioning (EP)

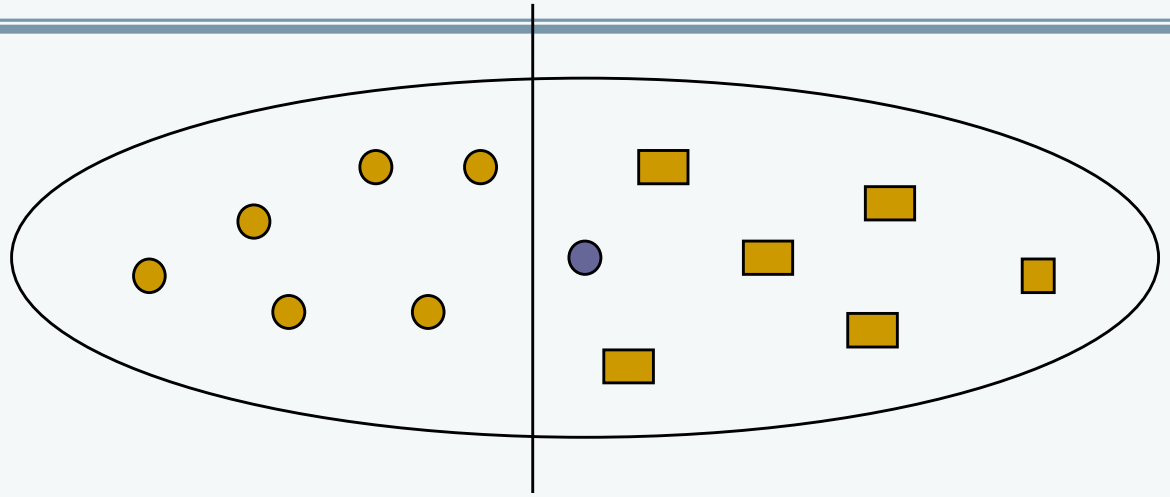
Invalid partition	Valid (for 3% interest)		Valid (for 5%)		Valid (for 7%)
–$0.01	$0.00	$100.00	$100.01	$999.99	$1,000.00

- Sample balances to calculate interest: –$10.00, $55.00, $270.00, and $1,260.00 → % of coverage?
- Compare to a naïve tester who would test every $50 such as $50.00, $100.0, $150.00, $200.00, … $800.00 (when they get tired) → % of coverage?

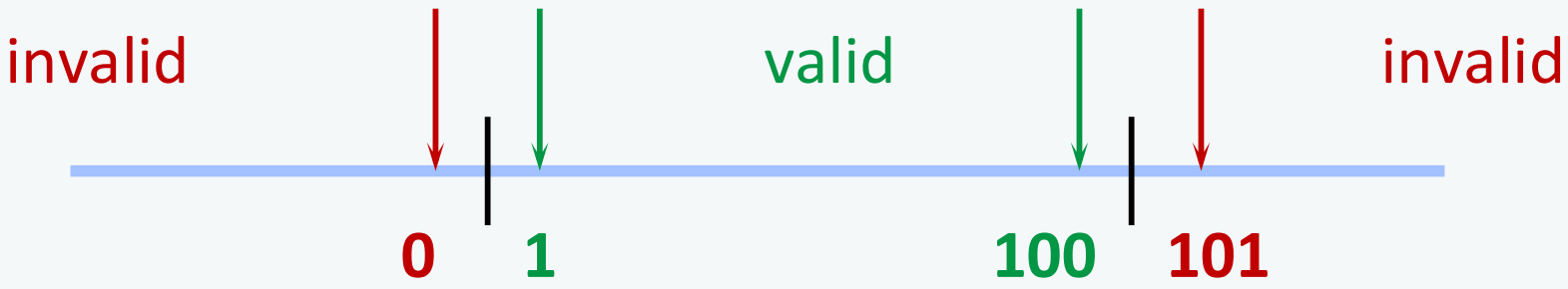
Equivalence Partitioning (EP)

- Equivalence partitions is aka Equivalence classes
- Partition that contains valid values is called a "valid equivalence partition", and invalid values is called an "invalid equivalence partition"
- Each value must belong to one and only one equivalence partition
- Any partition may be divided into sub partitions if required
- Coverage is measured as the number of equivalence partitions tested by at least one value, divided by the total number of identified equivalence partitions
- When invalid equivalence partitions are used, they should be tested individually. Whereas, values from valid partitions can be combined

Boundary value analysis (BVA)



- Faults tend to lurk near boundaries
- Good place to look for faults
- Test values on both sides of boundaries



Boundary value analysis (BVA)

- Boundary value analysis (BVA) is an extension of EP but can only be used when the partition is ordered. The minimum and maximum values of a partition are its boundary values.
- Some variations of this technique identify three boundary values per boundary: the values before, at, and just over the boundary
- BVA can be applied at all test levels. This technique is generally used to test requirements that call for a range of numbers.
- Boundary coverage for a partition is expressed as a percentage:

$$\frac{\#boundary\ values\ tested}{\#\ identified\ boundary\ test\ values}$$

Example: Loan Application

Customer Name

Account number

Loan amount requested

Term of loan

Monthly repayment

Term:

Repayment:

Interest rate:

Total paid back:

2-64 chars.

6 digits, 1st non-zero

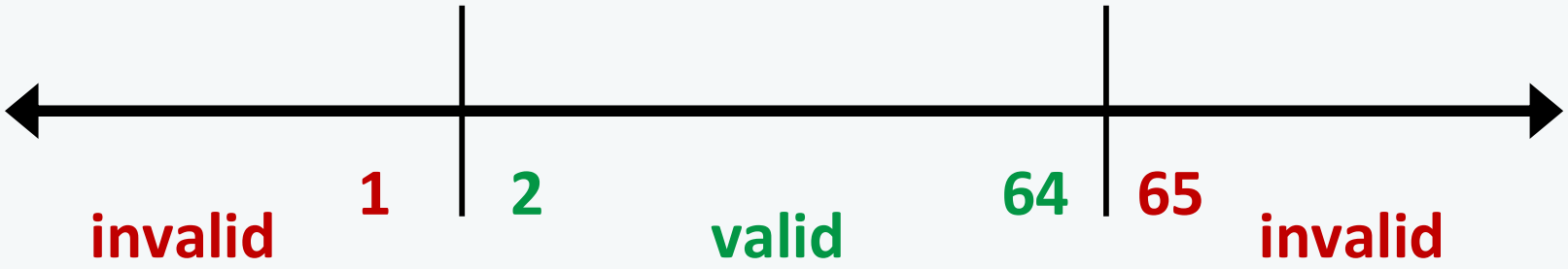
£500 to £9000

1 to 30 years

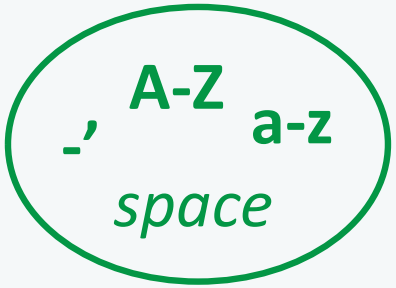
Minimum £10

Customer name

Number of characters:



Valid characters:



Any other

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Customer name	2 to 64 chars	< 2 chars	2 chars	1 char
		> 64 chars	64 chars	65 chars
	valid chars	invalid chars		0 char

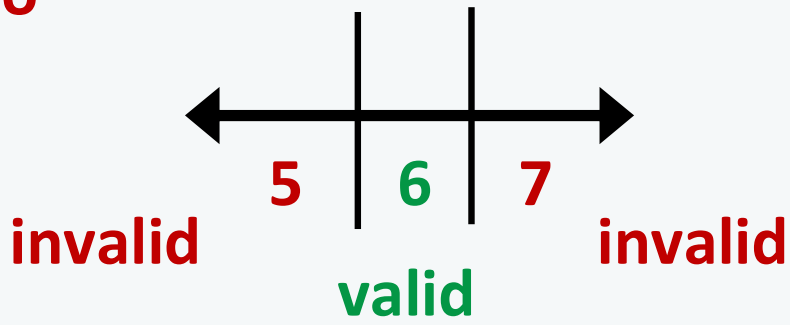
Account number

first character:

valid: non-zero

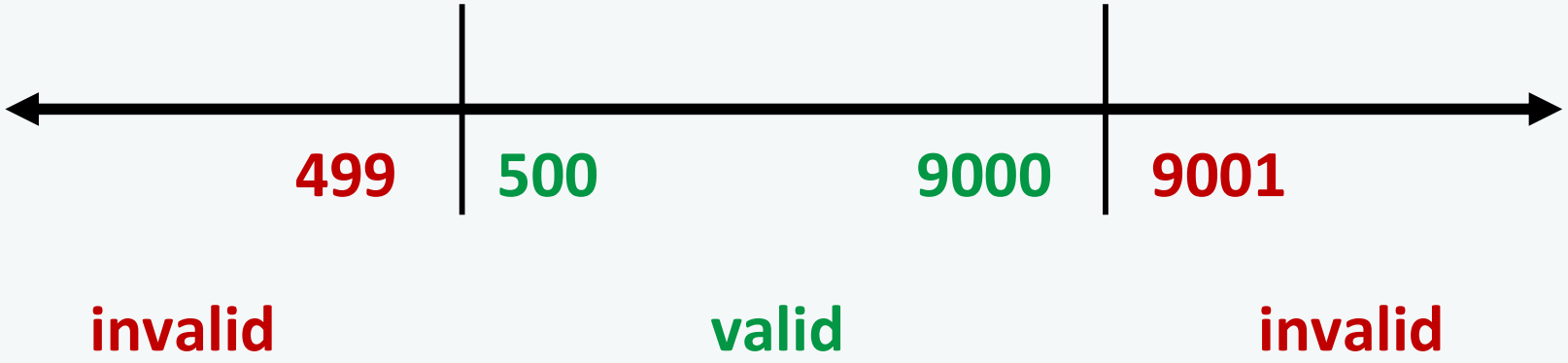
invalid: zero

number of digits:



Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Account number	6 digits	< 6 digits	100000	99999
		> 6 digits	999999	1000000
	1st non-zero	1st digit = 0		_ (0 digit)
		non-digit		

Loan amount



Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Loan amount	500 – 9000	< 500	500	499
		> 9000	9000	9001
		0		
		non-numeric		
		null		

Condition Template

Condition	Valid Partitions	Tag	Invalid Partitions	Tag	Valid Boundaries	Tag	Invalid Boundaries	Tag
Customer name	2 – 64 chars	VP1	< 2 chars	IP1	2 chars	VB1	1 char	IB1
	valid chars	VP2	> 64 chars	IP2	64 chars	VB2	65 chars	IB2
			invalid chars	IP3			0 char	IB3
Account number	6 digits	VP3	< 6 digits	IP4	100000	VB3	99999	IB4
	1st non-zero	VP4	> 6 digits	IP5	999999	VB4	1000000	IB5
			1st digit = 0	IP6			_ (0 digit)	IB6
			non-digit	IP7				
Loan amount	500 – 9000	VP5	< 500	IP8	500	VB5	499	IB7
			> 9000	IP9	9000	VB6	9001	IB8
			0	IP10				
			non-integer	IP11				
			null	IP12				

Design Test Cases

Test Case	Description	Expected Outcome	New Tag Covered
1	Name: John Smith Acc. No: 123456 Loan: 2500 Term: 3 years	Term: 3 years Repayment: 79.86 Interest rate: 10% Total paid: 2874.96	V1, V2, V3, V4, V5,…
2	Name: AB Acc. No: 100000 Loan: 500 Term: 1 year	Term: 1 year Repayment: 44.80 Interest rate: 7.5% Total paid: 537.60	B1, B3, B5…

Why do both EP and BVA?

- If you do boundaries only, you have covered all the partitions as well
  - technically correct and may be OK if everything works correctly!
  - if the test fails, is the whole partition wrong, or is a boundary in the wrong place - have to test mid-partition anyway
  - testing only extremes may not give confidence for typical use scenarios (especially for users)
  - boundaries may be harder (more costly) to set up

Test objectives?

Condition	Valid Partitions	Tag	Invalid Partitions	Tag	Valid Boundaries	Tag	Invalid Boundaries	Tag

- For a thorough approach: VP, IP, VB, IB
- Under time pressure, depends on your test objective
  - minimal user-confidence: VP only?
  - maximum fault finding: VB first (plus IB?)

Decision tables

- Explore combinations of inputs, situations or events,
- It is very easy to overlook specific combinations of input
- Start by expressing the input conditions of interest so that they are either TRUE or FALSE

  - record found
  - file exists
  - code valid
  - policy expired
  - account in credit
  - due date > current date

Example: Discount on SWT Course

A Software Testing course is created depending on this criteria:

- If the trainee is a **student**, he/she will get **20%** discount.
- If the trainee is **unemployed**, he/she will get **20%** discount.
- The trainee cannot be both **student** & **employed** at the same time.
- If the trainee is a student in (or a graduate from) **computer science** department, he/she will be directed to the **advanced** course group.

What are the input and output conditions?

List the Conditions & Actions

- List the **Conditions** of all inputs in the first column of the table
- List the **Actions/Outcomes** under the input conditions

Conditions	
Student	
Employed	
CS major	
Actions/Outcomes	
Basics 0% dis.	
Basics 20% dis.	
Adv. 0% dis.	
Adv. 20% dis.	

Identify Input Combinations

- Add columns to the table for each unique combination of input conditions.
- Each entry in the table may be either 'T' for true, 'F' for false.

Conditions	1	2	3	4	5	6	7	8
Student	T	T	T	T	F	F	F	F
Employed	T	T	F	F	T	T	F	F
CS major	T	F	T	F	T	F	T	F
Actions/Outcomes								
Basics 0% dis.								
Basics 20% dis.								
Adv. 0% dis.								
Adv. 20% dis.								

Rationalise Input Combinations

- Some combinations may be impossible or not of interest
- Some combinations may be 'equivalent'
- Use a hyphen to denote "don't care"

Rationalise Input Combinations

Conditions	1	2	3	4	5	6	7	8
Student	T	T	T	T	F	F	F	F
Employed	T	T	F	F	T	T	F	F
CS major	T	F	T	F	T	F	T	F
Actions/Outcomes								
Basics 0% dis.								
Basics 20% dis.								
Adv. 0% dis.								
Adv. 20% dis.								

Reduce # of Test Cases

Conditions	1	2	3	4	5	6	7	8
Student	T		T	T	F	F	F	F
Employed	T		F	F	T	T	F	F
CS major	–		T	F	T	F	T	F
Actions/Outcomes								
Basics 0% dis.								
Basics 20% dis.								
Adv. 0% dis.								
Adv. 20% dis.								

Reduce # of Test Cases

Conditions	1	2	3	4	5	6	7	8
Student	T		–	T	F	F		F
Employed	T		F	F	T	T		F
CS major	–		T	F	T	F		F
Actions/Outcomes								
Basics 0% dis.								
Basics 20% dis.								
Adv. 0% dis.								
Adv. 20% dis.								

Reduce # of Test Cases

Conditions	1	2	3	4	5	6	7	8
Student	T		–	–	F	F		
Employed	T		F	F	T	T		
CS major	–		T	F	T	F		
Actions/Outcomes								
Basics 0% dis.								
Basics 20% dis.								
Adv. 0% dis.								
Adv. 20% dis.								

Reduce # of Test Cases

Conditions	1	3	4	5	6	2	7	8
Student	T	T	T	F	F			
Employed	T	F	F	T	T			
CS major	T	T	F	T	F			
Actions/Outcomes								
Basics 0% dis.								
Basics 20% dis.								
Adv. 0% dis.								
Adv. 20% dis.								

Rationalising outputs

- if outputs or effects are mutually exclusive, I.e. T occurs in only one place in each column, we can combine them
- for example:

X	T	F	F
Y	F	T	F
Z	F	F	T

is equivalent to:

Output	X	Y	Z

Rationalising dangers

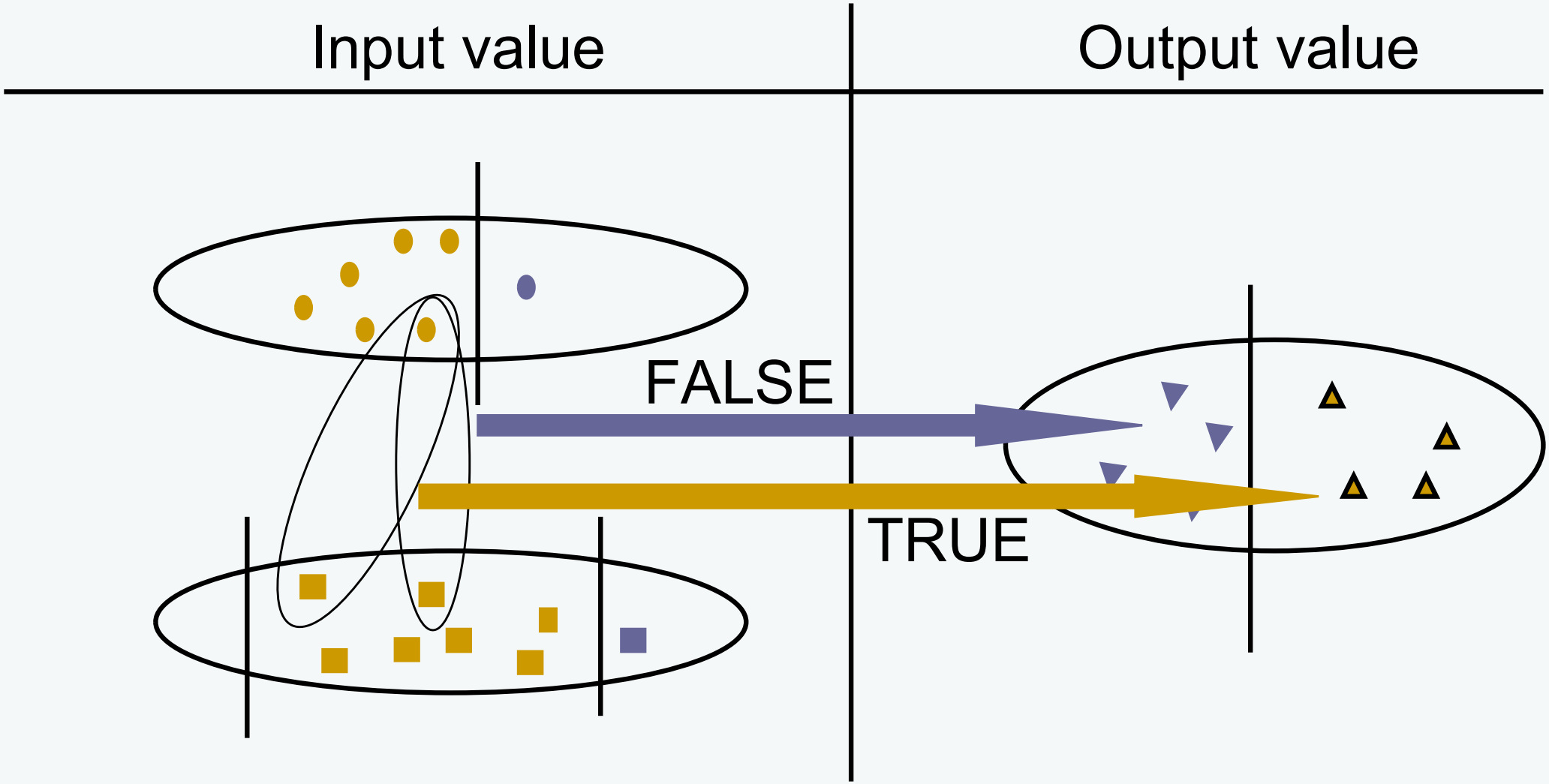
- Rationalising is based on assumptions
- Assumptions may be wrong!
- Assumptions should be stated
- Assumptions may change over time
- Be aware of the dangers
  - Filling in the full table may find errors which will be missed if you rationalise
  - It is possible to rationalise too far

Extending decision tables

- Entries can be more than just ‘true’ or ‘false’
  - completing table needs to be done carefully
  - rationalising becomes more important
- E.g.

Code = 1, 2, or 3	1	1	1	1	2	2	2	2	3	3	3	3
Exp.date < now	T	T	F	F	T	T	F	F	T	T	F	F
Class A product	T	F	T	F	T	F	T	F	T	F	T	F

Decision Tables in relation to EP and BVA



State Transition Testing

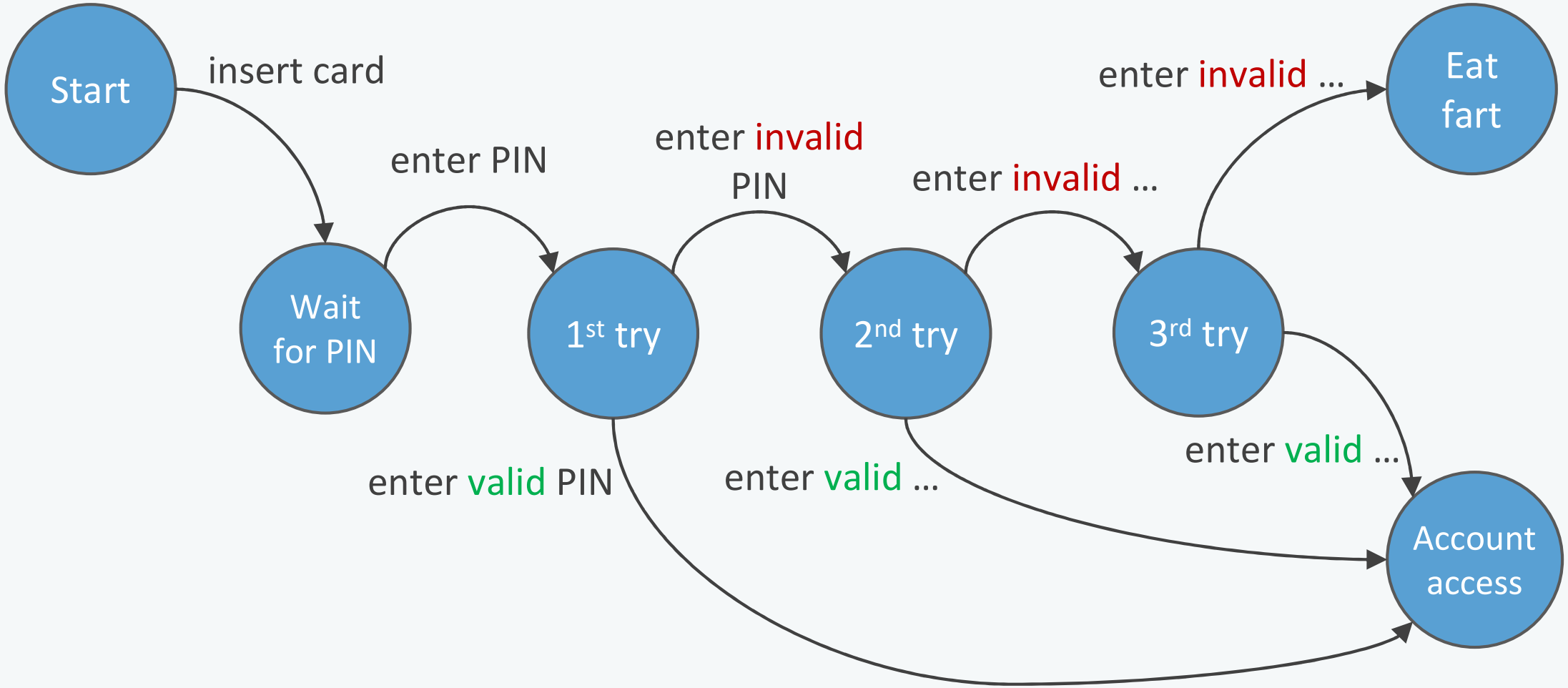
- A state transition diagram shows a limited (finite) number of different states, and the transitions from one state to another a determined by the rules of the 'machine'.
- A transition is initiated by an event (e.g., user input of a value into a field). The event results in a transition.
- The state change may result in the software taking an action (e.g., outputting a calculation or error message).
- A state transition table shows all valid transitions and potentially invalid transitions between states.
- State transition diagrams normally show only the valid transitions and exclude the invalid transitions.

State Transition Testing

- State transition testing is a black-box test technique in which test cases are designed to exercise elements of a state transition model.
- Tests can be designed to cover a typical sequence of states, to exercise all states, to exercise every transition, to exercise specific sequences of transitions, or to test invalid transitions.
- Coverage is normally expressed as a percentage

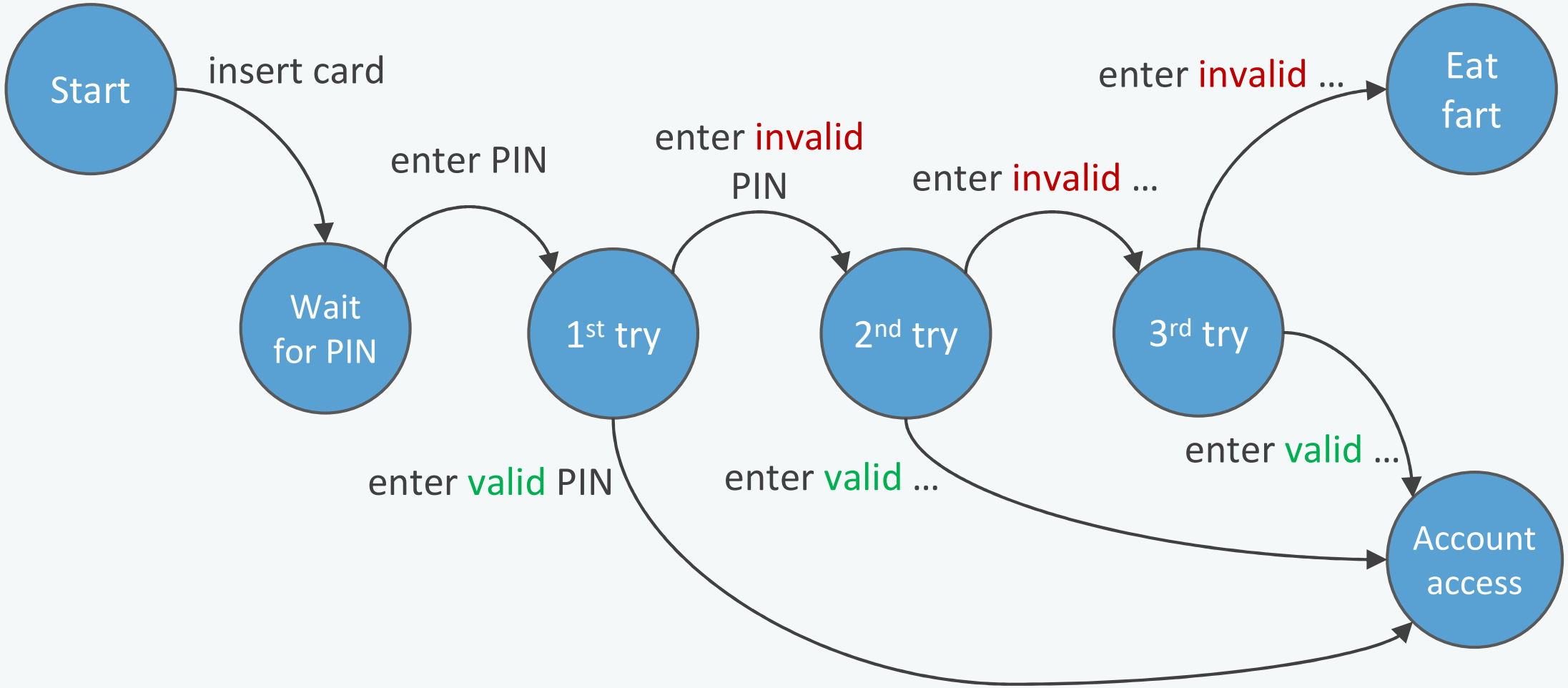
$$\frac{\#tested\ states/transitions}{\#\ identified\ states/transitions}$$

State Transition Testing



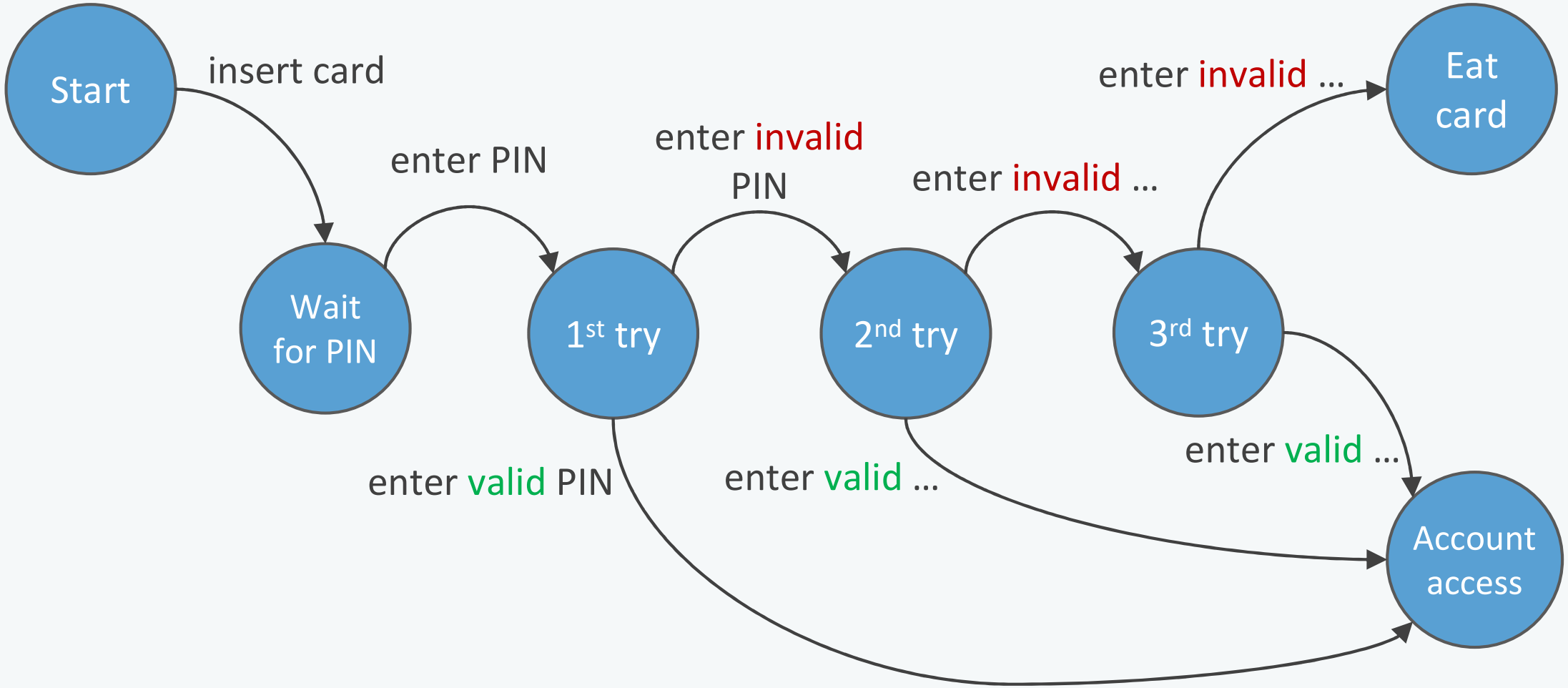
test required to cover all states?

State Transition Testing



test required to cover all states? **2**

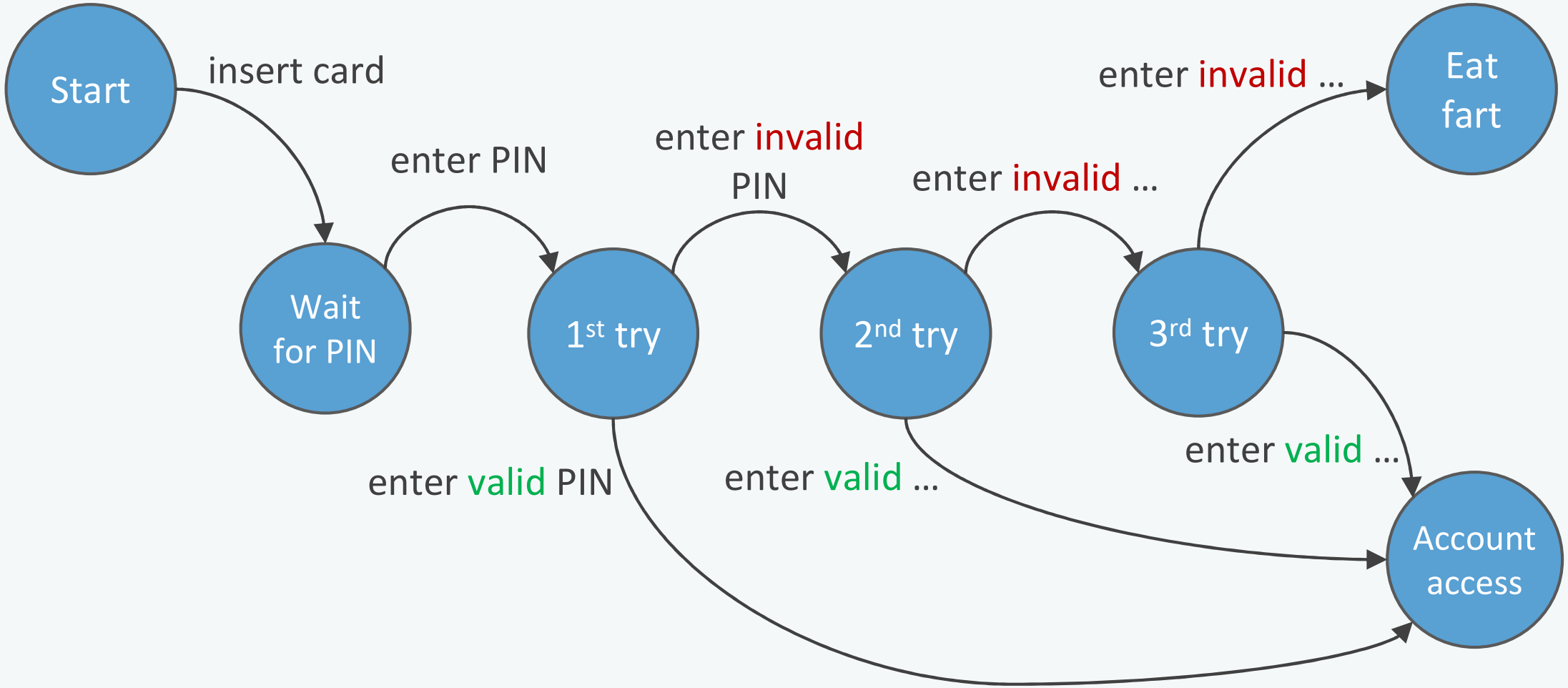
State Transition Testing



test required to cover all states? **2**

test required to cover all transitions?

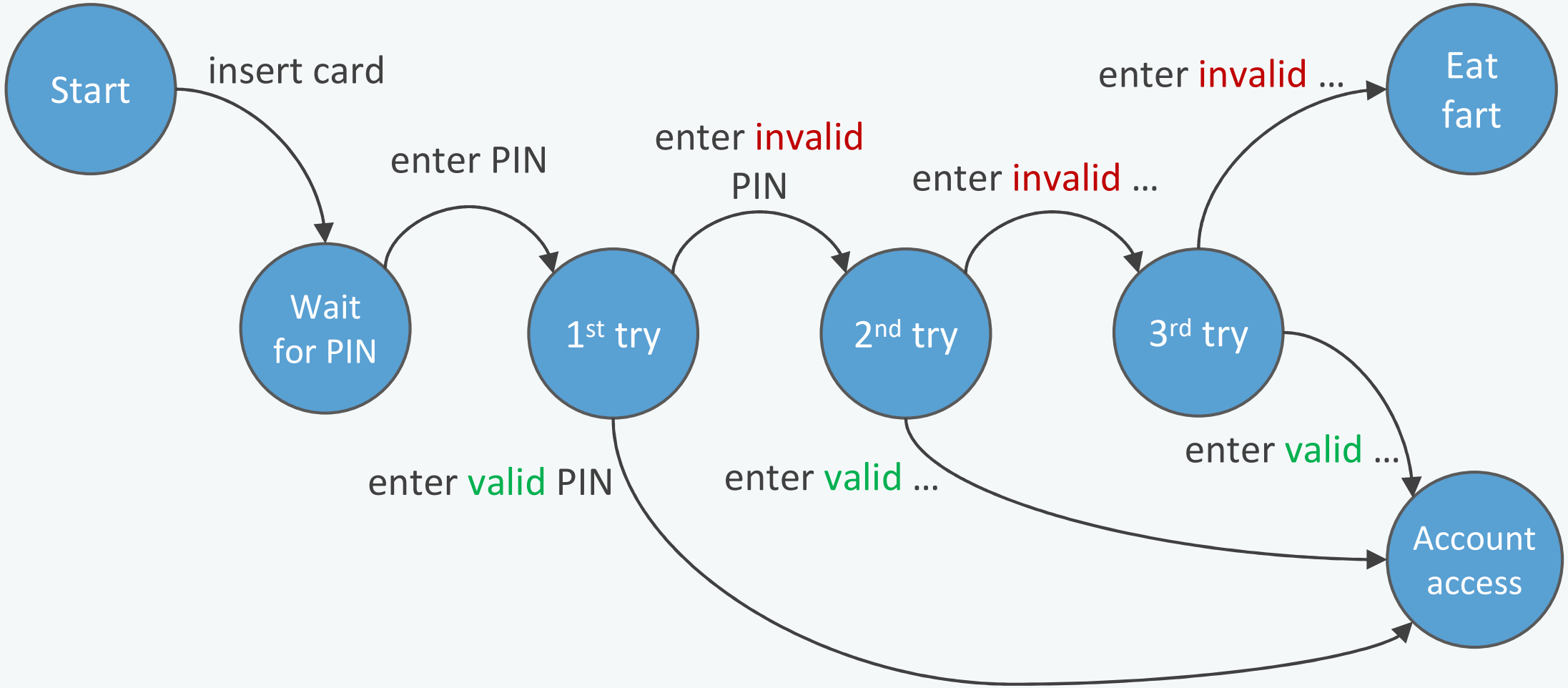
State Transition Testing



test required to cover all states? **2**

test required to cover all transitions?

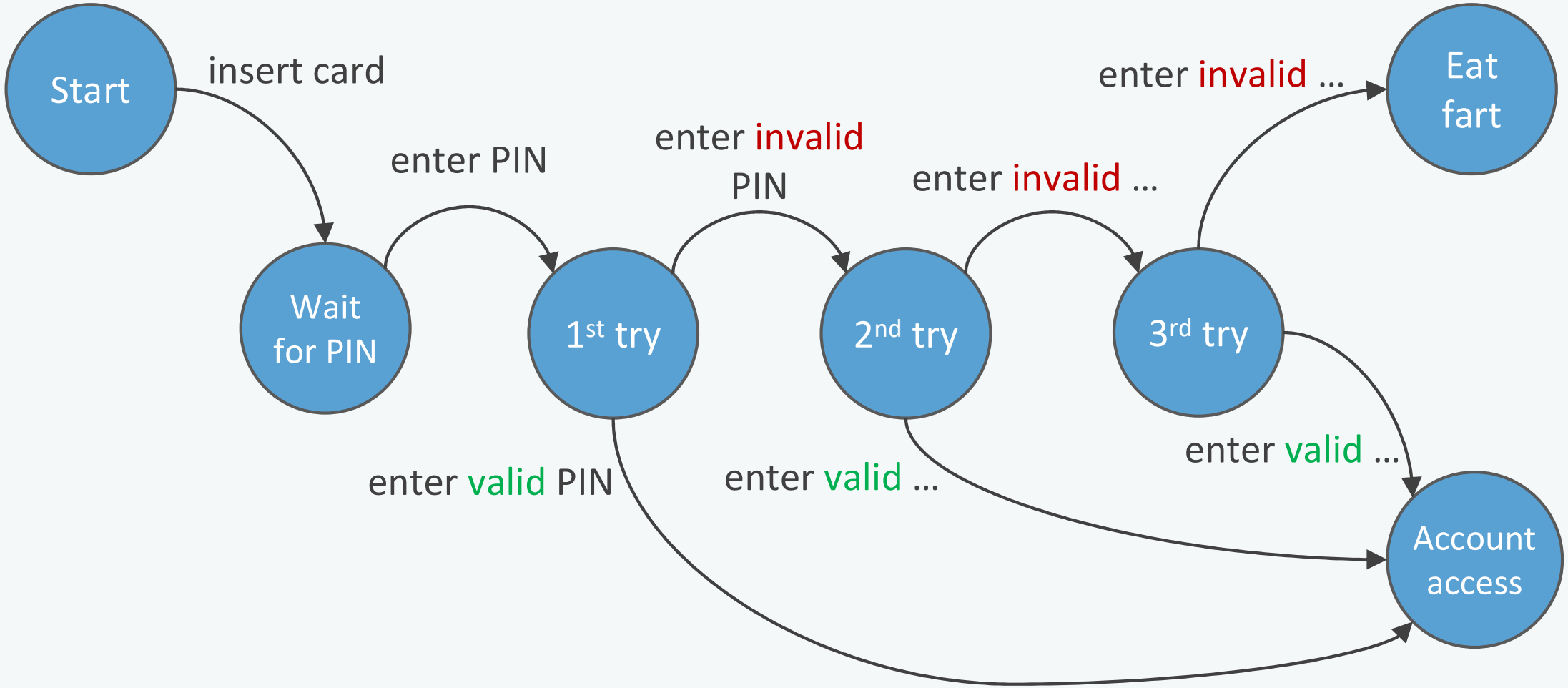
State Transition Testing



test required to cover all states? **2**

test required to cover all transitions?

State Transition Testing



test required to cover all states? **2**

test required to cover all transitions? **4**

State Transition Testing

invalid transition

Events

States

	Insert card	Valid PIN	Invalid PIN
S1) Start state	S2	?	–
S2) Wait for PIN	?	S6	S3
S3) 1st try invalid	?	S6	S4
S4) 2nd try invalid	–	S6	S5
S5) 3rd try invalid	–	–	S7
S6) Access account	–	– / ?	– / ?
S7) Eat card	S1 (new card)	–	–

Question

You are testing an unattended gasoline pump that only accept credit cards. Once the card is validated, the pump nozzle placed into the tank, and the desired grade selected, the customer enters the desired amount of fuel in gallons with the keypad. The keypad only allows the entry of digits. Fuel is sold in tenths (0.1) of a gallon, up to 50.0 gallons.

Which of the following is a minimum set of desired amounts that covers the equivalence partitions for this input?

A. 0.0, 10.0, 60.0

B. 0.0, 0.1, 50.0

C. 0.0, 0.1, 50.0, 70.0

D. -0.1, 0.0, 0.1, 49.9, 50.0, 50.1

Question

You are testing a machine that scores exam papers and assign grades. Based on the score achieved, the grades are as follows:

1 – 49 = F; 50 – 59 = D-; 60 – 69 = D;

70 – 79 = C; 80 – 89 = B; 90 – 100 = A

If you apply EP, how many test cases will you need to achieve minimum test coverage?

A. 6

B. 8

C. 10

D. 12

Question

You are testing a machine that scores exam papers and assign grades. Based on the score achieved, the grades are as follows:

1 – 49 = F; 50 – 59 = D-; 60 – 69 = D;

70 – 79 = C; 80 – 89 = B; 90 – 100 = A

If you apply BVA, how many test cases will you need to achieve minimum test coverage?

A. 8

B. 10

C. 12

D. 14

Question

You are testing a scale system that determines shipping rates for a web-based auto part distributor. Identify how many equivalence classes are necessary for the following range. Weights are rounded to the nearest pound.

Weight	1 … 10 lbs	11 … 25 lbs	26 … 50 lbs	51 lbs & up
Shipping cost	$5.00	$7.50	$12.00	$17.00

A. 8

B. 6

C. 5

D. 4

Question

Which of the following statements are true for the EP test technique?

Divides possible inputs into classes that have the same behaviour

Uses both valid and invalid partitions

Makes use only of valid partitions

Must include at least two values from every equivalence partition

Can be used only for testing systems that have a GUI

a) A, B & E

b) A, C & D

c) A & E

d) A & B

Question

An employee's bonus is to be calculated. It cannot be negative, but it can be calculated down to zero. The bonus is based on the length of employment. An employee can be employed for less than or equal to 2 years, more than 2 years but less than 5 years, 5 to 10 years, or longer than 10 years.

What is the minimum number of test cases required to cover all valid equivalence partitions for calculating the bonus?

A. 3

B. 5

C. 2

D. 4

Question

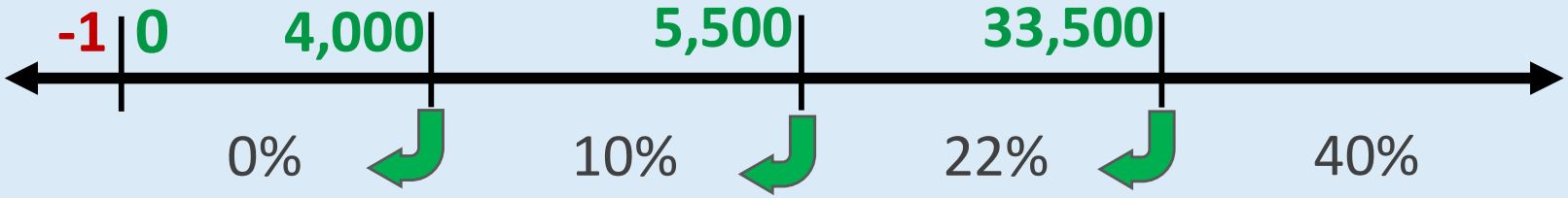
In a system designed to work out the tax to be paid, an employee has $4,000 of salary tax free. The next $1,500 is taxed at 10%. The next $28,000 is taxed at 22%. Any further amount is taxed at 40$. Which of these groups of numbers would fall into the same equivalence class?

A. 4,800 – 14,000 – 28,000

B. 5,200 – 5,500 – 28,000

C. 28,001 – 32,000 – 35,000

D. 5,800 – 28,000 – 32,000



Question

A smart home app measures the average temperature in the house over the previous week & provide feedback to the occupants on their environmental friendliness based on this temperature. The feedback is as such:

- Up to 14°C – Freezing
- 15°C to 19°C – Cold
- 20°C to 24°C – Cool
- 25°C to 30°C – Open the window
- Above 30°C – Turn the AC on

Using BVA (only min- and max- value), which of the following sets of test inputs provides the highest level of boundary coverage?

A. 0°C, 15°C, 25°C, 28°C, 29°C

B. 13°C, 19°C, 24°C, 30°C, 50°C

C. 14°C, 20°C, 24°C, 30°C, 31°C

D. 18°C, 19°C, 23°C, 24°C, 29°C, 30°C

Question

An e-commerce system sells cooking supplies in bulk. The units in which the items are sold are either grams (expensive items) or kilograms (cheap items). Regardless of the units, the smallest valid order amount is 0.5 units and the largest valid order amount is 25.0 units. The precision of the units field is 0.1 units.

Which of the following is a set of input values that cover the boundary values with 2-point boundary values for this field?

A. 0.3, 10.0, 28.0

B. 0.4, 0.5, 0.6, 14.9, 25.0, 25.1

C. 0.4, 0.5, 25.0, 25.1

D. 0.5, 0.6, 24.9, 25.0

Question

You're testing a scale system that determines shipping rates for a regional web-based auto parts distributor. Due to regulations, shipments cannot exceed 100 lbs. You want to include boundary value analysis as part of your black-box test design. How many test will you need to execute to achieve 100% BVA?

Weight	1 … 10 lbs	11 … 25 lbs	26 … 50 lbs	51 … 100 lbs
Shipping cost	$5.00	$7.50	$12.00	$17.00

A. 4

B. 8

C. 10

D. 12

Question

Consider the following decision table for the portion of an online airline reservation system that allows frequent flyers to redeem points for reward travel:

Condition	1	2	3
Account/password OK	F	T	T
Sufficient points	-	F	T
Action			
Show flight history	F	T	T
Allow reward travel	F	F	T

There are 2 equivalence partitions for the condition where Account/Password OK is not True (i.e., False): (1) account is invalid, and (2) account is valid but password is invalid. There is only 1 equivalence partition for the condition where Account/Password OK is True: the account & password are valid. If you want to design tests to cover the equivalence partitions for Account/Password OK and also for this portion of the decision table, what is the minimum number of tests required?

4

CONTENTS

- Categories of Test Techniques
- Black-box Test Techniques
- White-box Test Techniques
  - Statement coverage
  - Decision coverage
- Experience-based Test Techniques

White-box Test Techniques

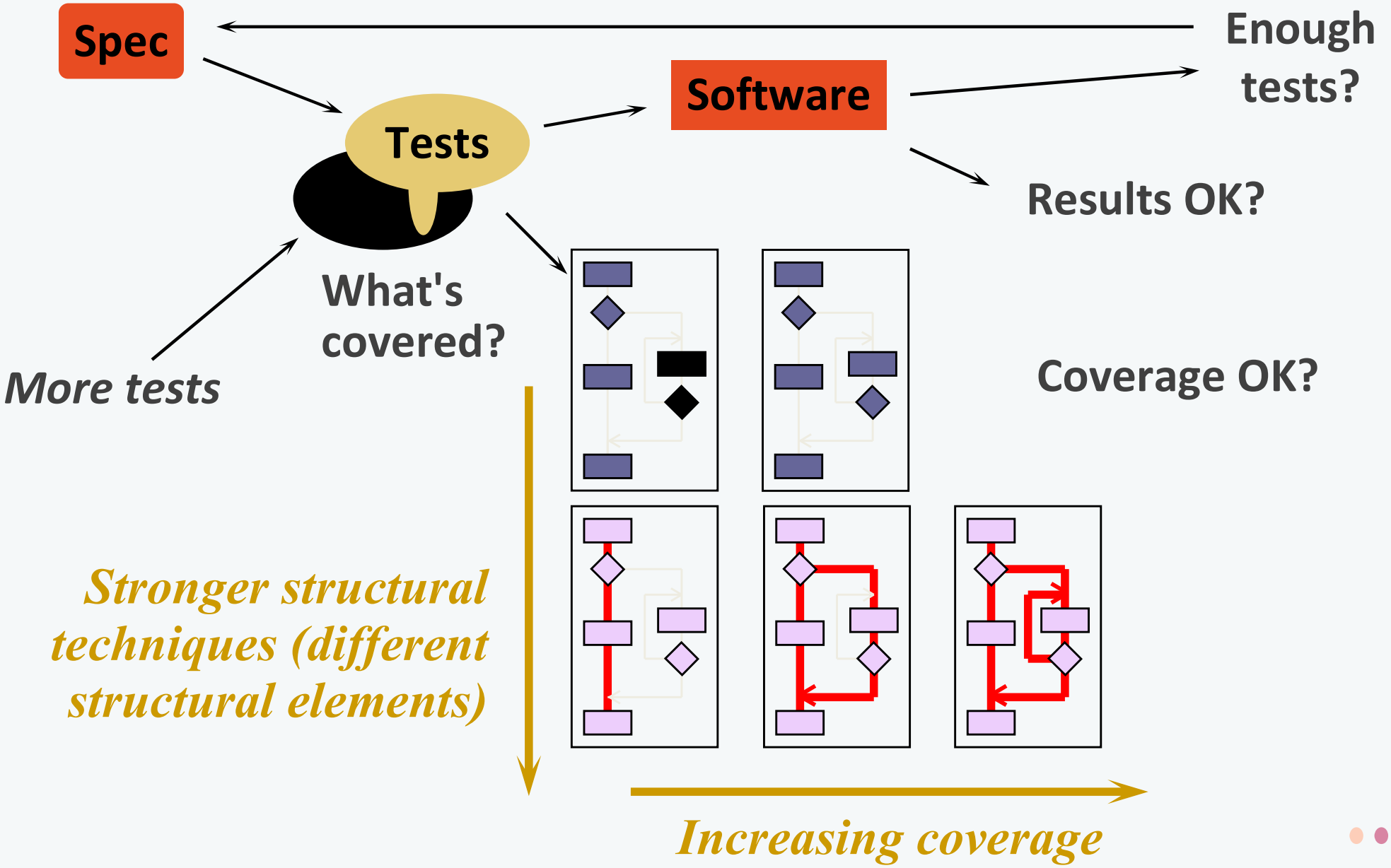
White-box test techniques serve 2 purposes: coverage measure & structural test case design

- Access the amount of testing performed by black-box tests – access coverage
- Design additional tests to increase the coverage

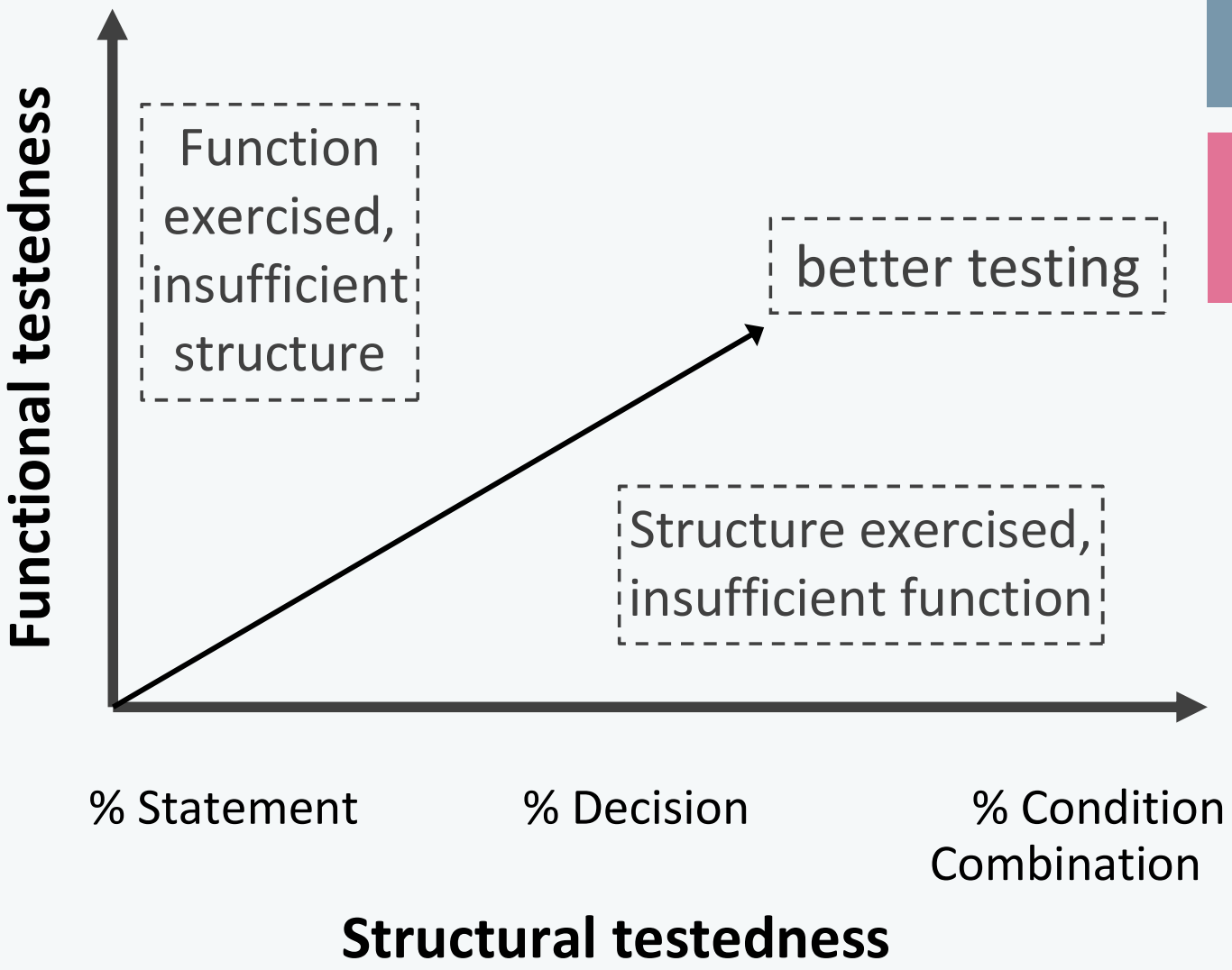
$$Coverage = \frac{\#\ coverage\ items\ exercised}{Total\ \#\ coverage\ items} \times 100\%$$

$coverage\ items$ = what we've been able to count & see whether a test has exercised or used this item.

Using Structural Coverage



The test coverage trap



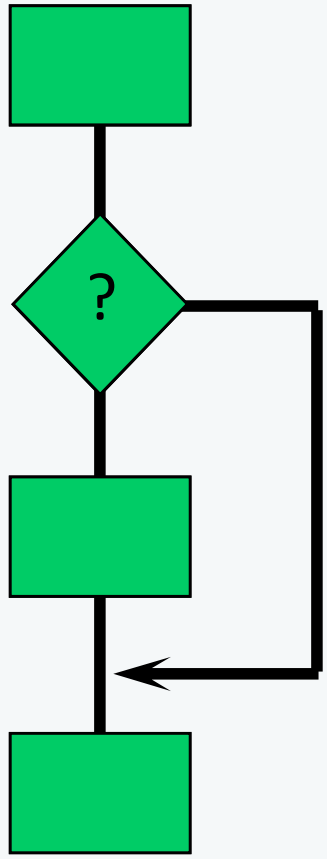
100% coverage does not mean 100% tested!

Coverage is only one aspect of thoroughness

Statement coverage

Percentage of executable statements exercised by a test suite

$$= \frac{number\ of\ statements\ exercised}{total\ number\ of\ statements}$$



Example:

- Program has 100 statements
- Tests exercise 87 statements
- Statement coverage = 87%

Statement coverage is normally measured by a software tool.

Typical ad hoc testing achieves 60 – 75%

Example of statement coverage

```
1  read(a)
2  IF a > 6 THEN
3      b = a
4  ENDIF
5  print b
```

Test case	Input	Expected output
1	7	7

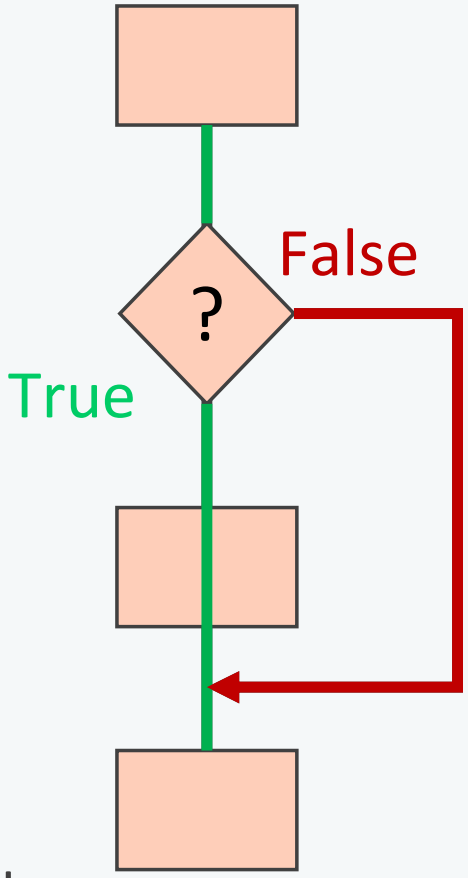
Statement numbers

As all 5 statements are 'covered' by this test case, we have achieved 100% statement coverage

Decision coverage (Branch coverage)

Percentage of decision outcomes exercised by a test suite

$$= \frac{number\ of\ decision\ outcomes\ exercised}{total\ number\ of\ decision\ outcomes}$$



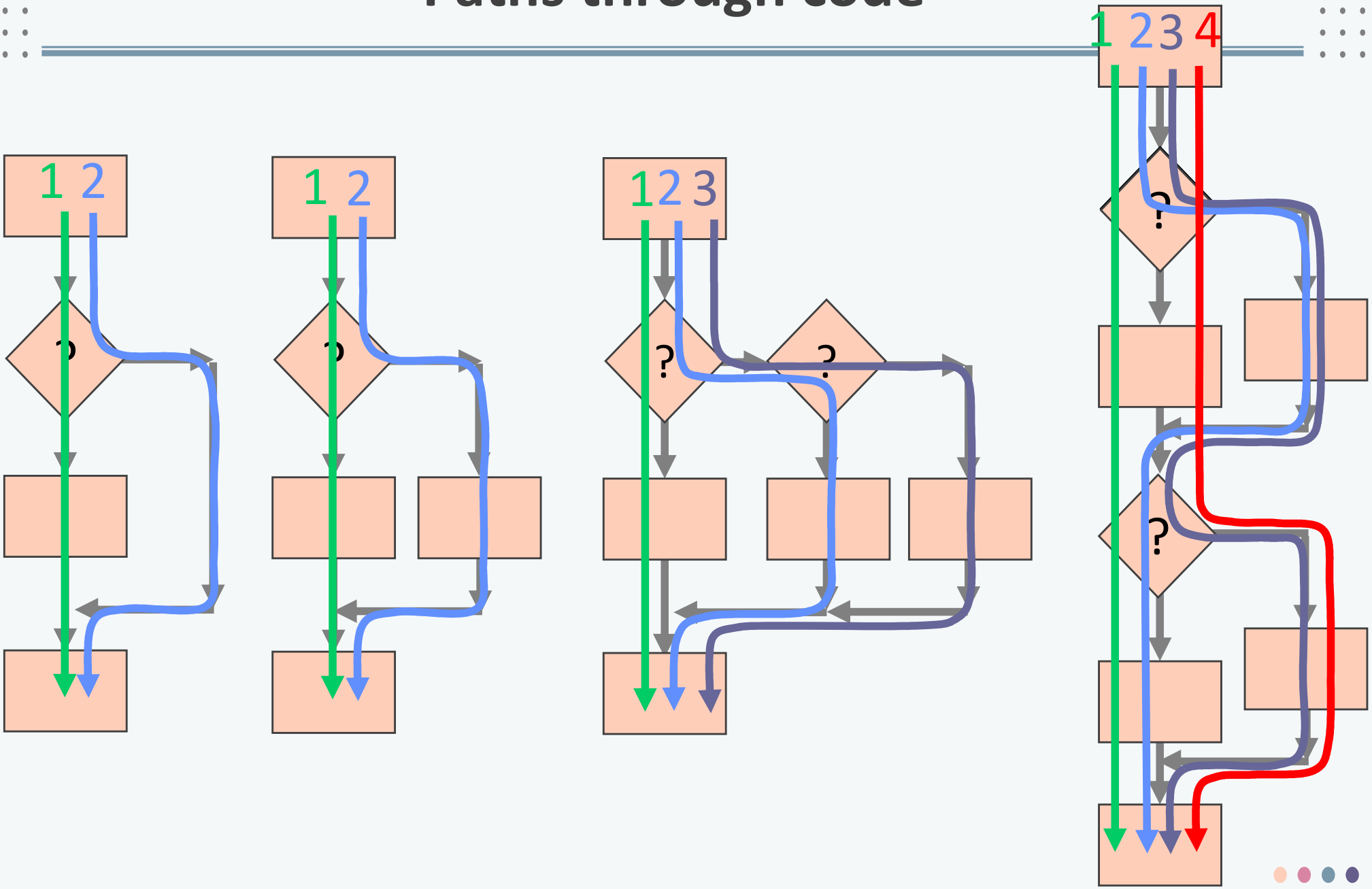
Example:

- Program has 120 statements
- Tests exercise 60 statements
- Statement coverage = 50%

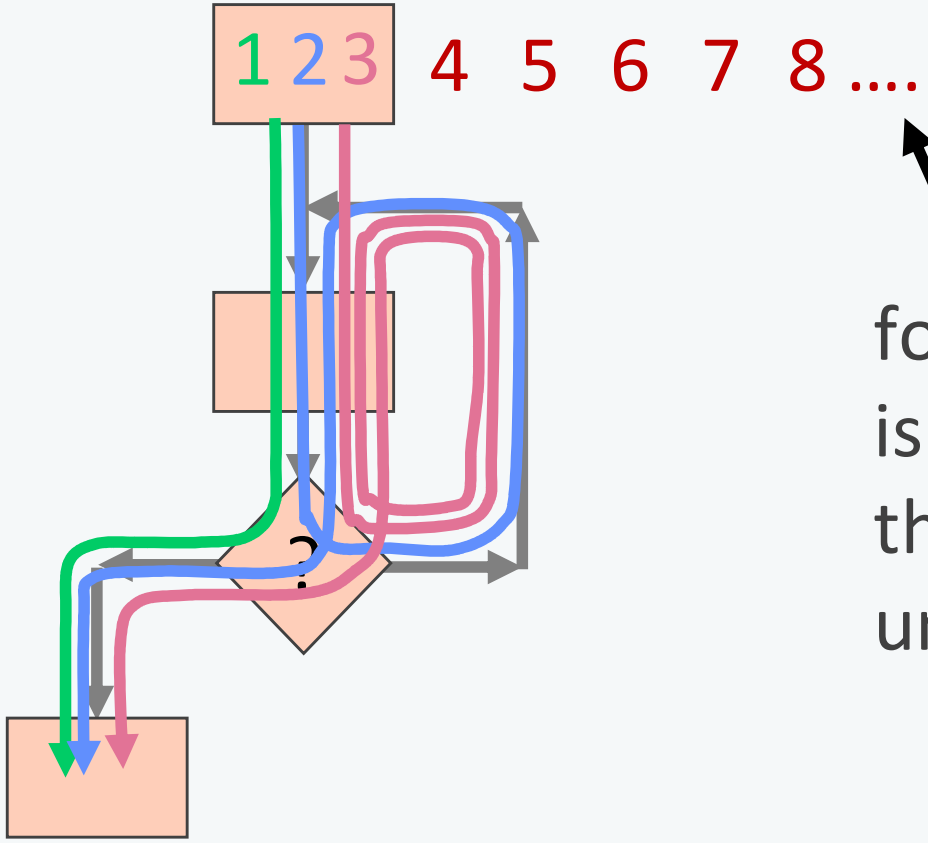
Decision coverage is normally measured by a software tool.

Typical ad hoc testing achieves 40 – 60%

Paths through code



Paths through code with loops



for as many times as it is possible to go round the loop (this can be unlimited, i.e. infinite)

Example 1

```
Wait for card to be inserted
IF card is a valid card THEN
  display "Enter PIN number"
  IF PIN is valid THEN
        select transaction
  ELSE (otherwise)
        display "PIN invalid"
ELSE (otherwise)
  reject card
End
```

Example 1

Wait for card to be inserted

IF card is a valid card THEN

  display "Enter PIN number"

  IF PIN is valid THEN

    select transaction

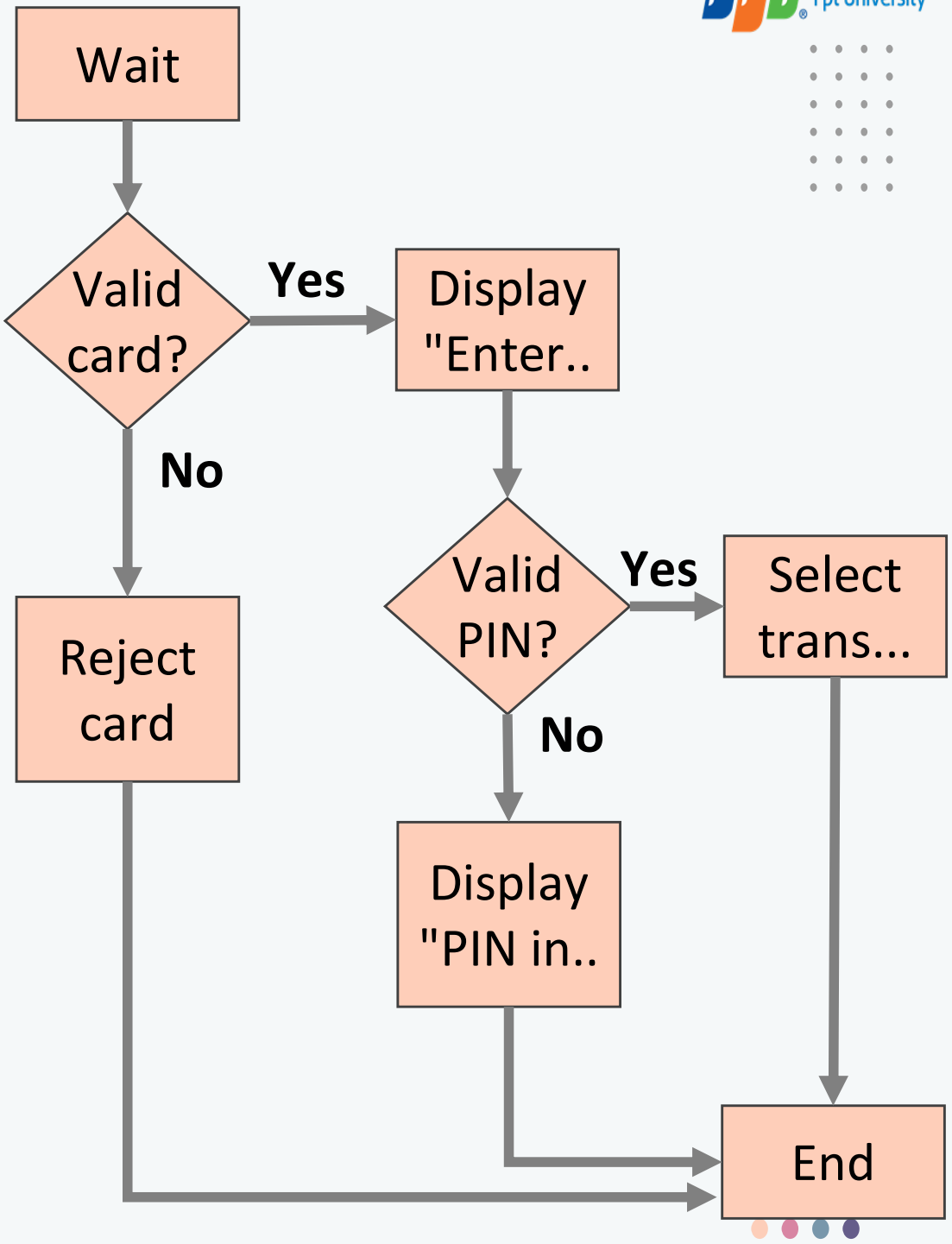
  ELSE (otherwise)

    display “PIN invalid”

ELSE (otherwise)

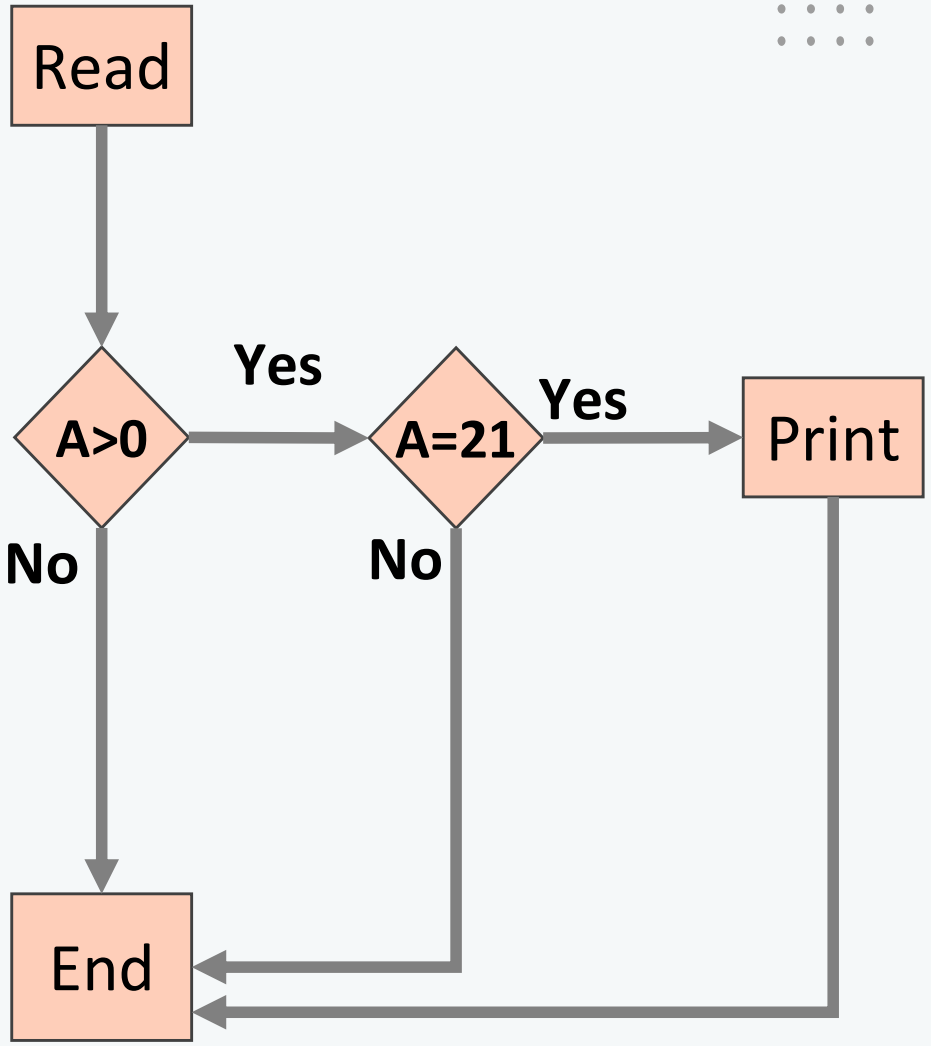
  reject card

End



Example 2

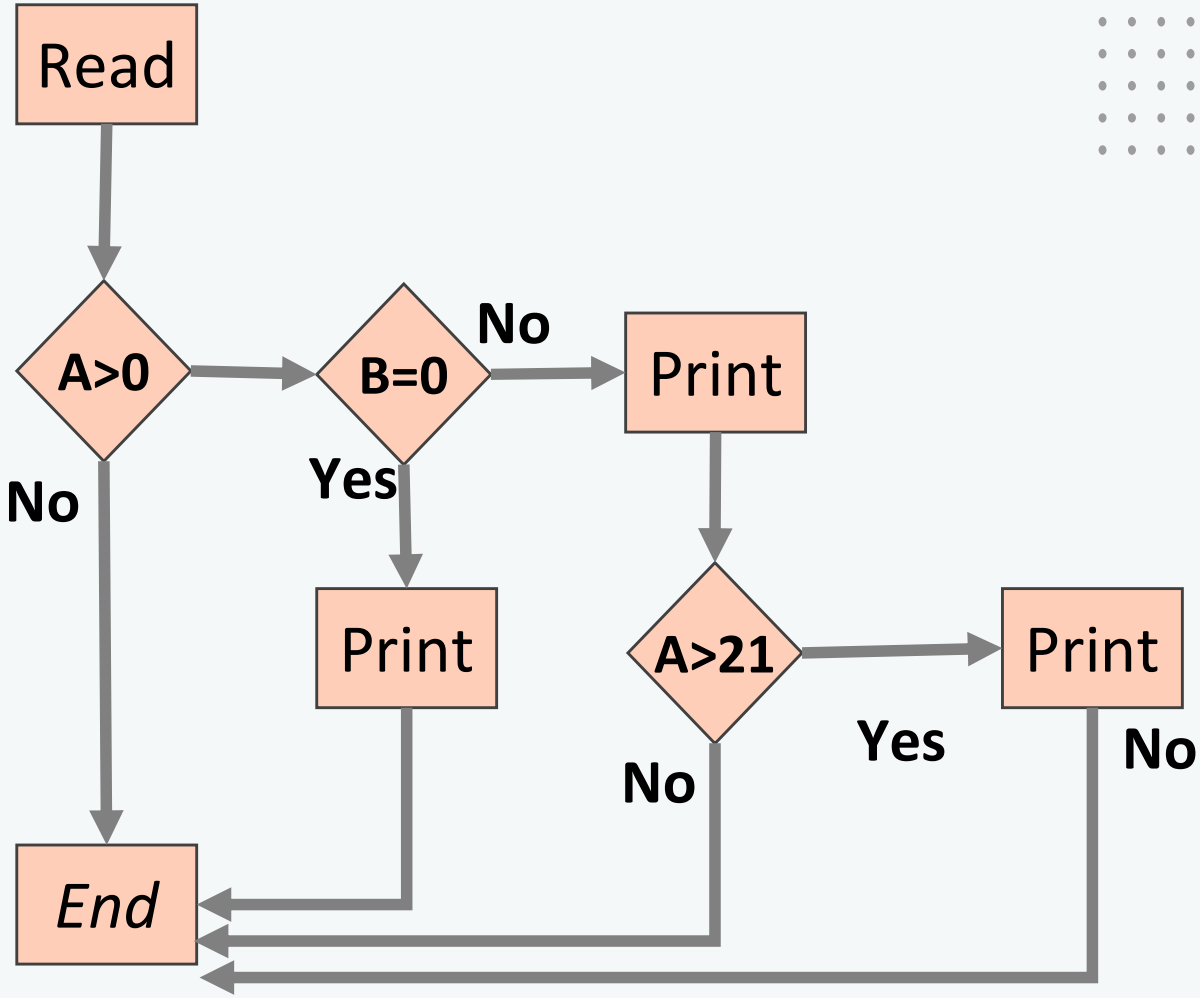
```
Read A
IF A > 0 THEN
   IF A  = 21 THEN
      Print "Key"
   ENDIF
ENDIF
```



- Cyclomatic complexity: 3
- Minimum tests to achieve:
  - Statement coverage: 1
  - Branch coverage: 3

Example 3

```
Read A
Read B
IF A > 0 THEN
   IF B  = 0 THEN
      Print "No values"
   ELSE
         Print B
         IF A > 21 THEN
            Print A
         ENDIF
   ENDIF
ENDIF
```

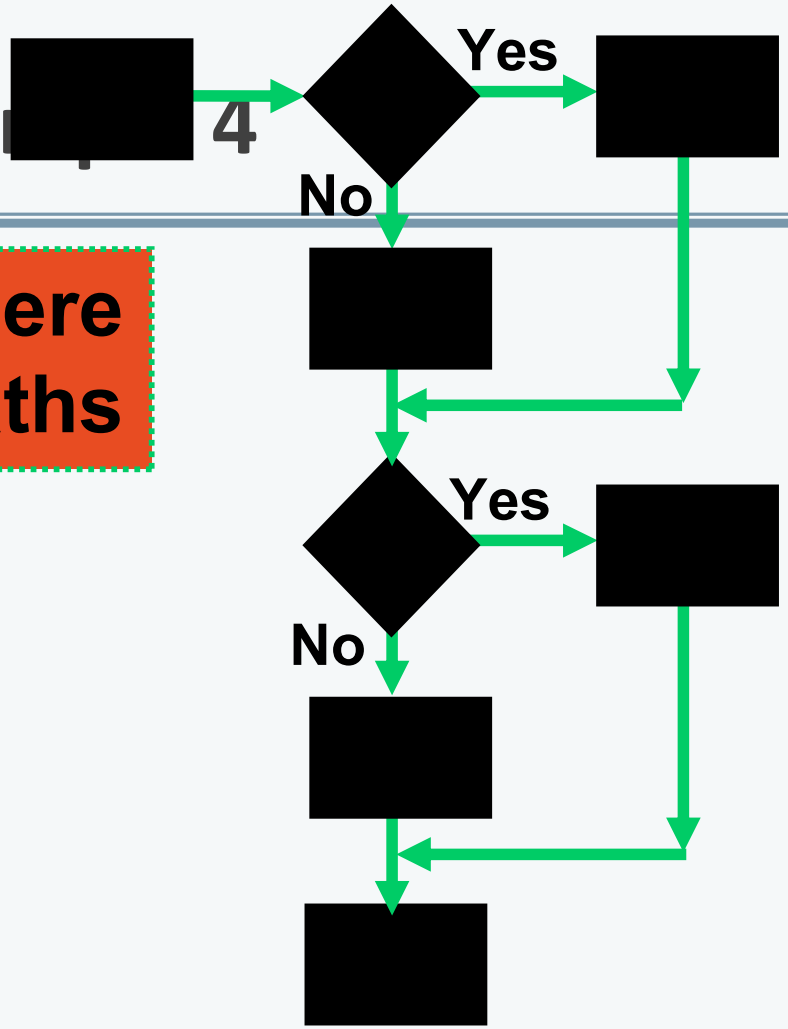


- Cyclomatic complexity: 4
- Minimum tests to achieve:
  - Statement coverage: 2
  - Branch coverage: 4

Exa[redacted] 4

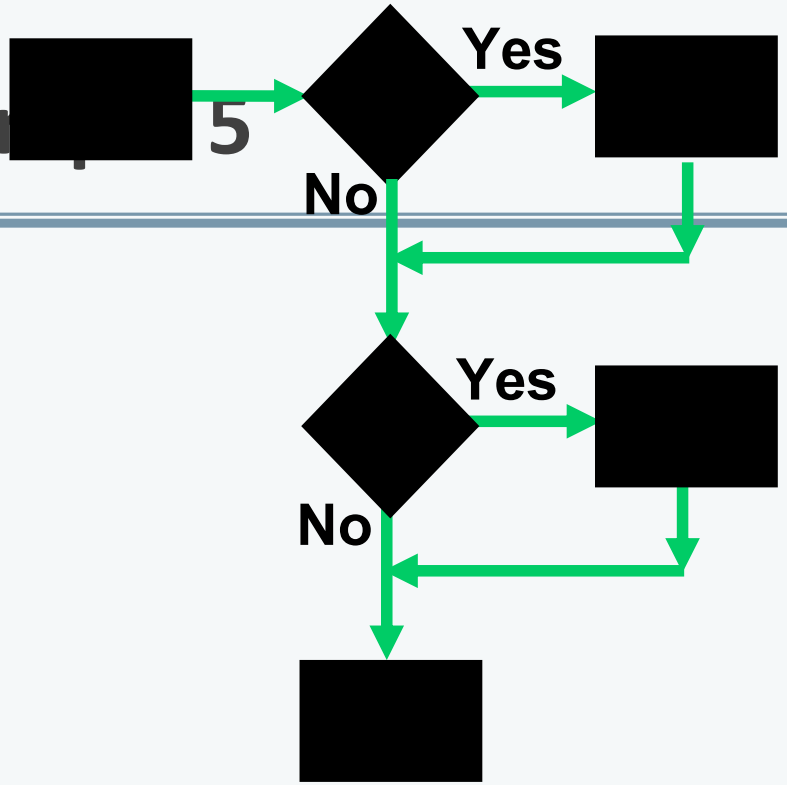
```
Read A
Read B
IF A < 0 THEN
    Print  “A negative”
ELSE
    Print  “A positive”
ENDIF
IF B < 0 THEN
    Print  “B negative”
ELSE
    Print  “B positive”
ENDIF
```

Note: there are 4 paths



- Cyclomatic complexity: _____ 3
- Minimum tests to achieve:
  - Statement coverage: _____ 2
  - Branch coverage: _____ 2

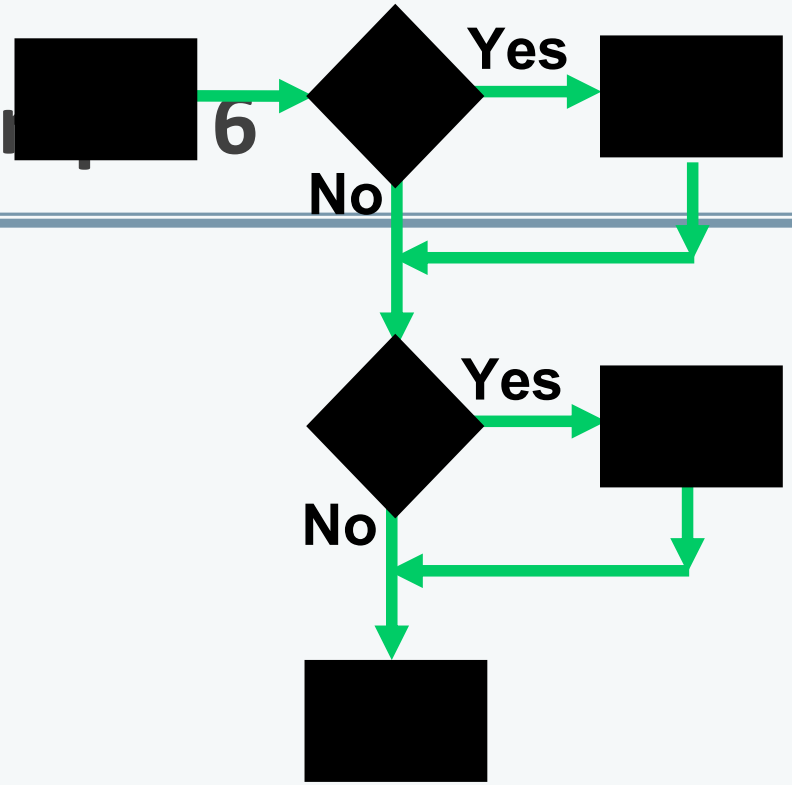
Exa[redacted]5




```
Read A
Read B
IF A < 0 THEN
    Print  “A negative”
ENDIF
IF B < 0 THEN
    Print  “B negative”
ENDIF
```

- Cyclomatic complexity: _____ 3
- Minimum tests to achieve:
  - Statement coverage: _____ 1
  - Branch coverage: _____ 2

Exa[redacted] 6




```
Read A
IF A < 0 THEN
    Print  "A negative"
ENDIF
IF A > 0 THEN
    Print  "A positive"
ENDIF
```

- Cyclomatic complexity: _____ 3
- Minimum tests to achieve:
  - Statement coverage: _____ 2
  - Branch coverage: _____ 2

CONTENTS

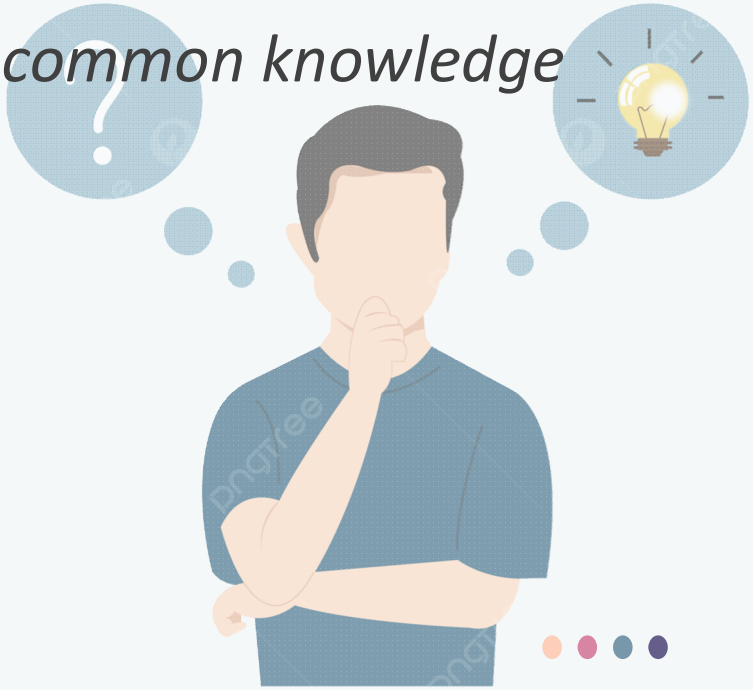
- Categories of Test Techniques
- Black-box Test Techniques
- White-box Test Techniques
- Experience-based Test Techniques

Experience-based techniques

- Test cases are derived from the tester's skill and intuition, and their experience with similar applications and technologies.
- Can be helpful in identifying tests that were not easily identified by other more systematic techniques.
- Depending on tester's approach and experience, these techniques may achieve widely varying degrees of coverage and effectiveness.
- Coverage can be difficult to assess and may be unmeasurable with these techniques.

Error Guessing

- A methodical approach to the error guessing technique is to create a list of possible mistakes, defects & failures → design tests that will expose those failures & the defects that caused them.
- These mistake, defect, failure lists can be built based on *experience*, *defect and failure data*, or from *common knowledge* about why software fails.

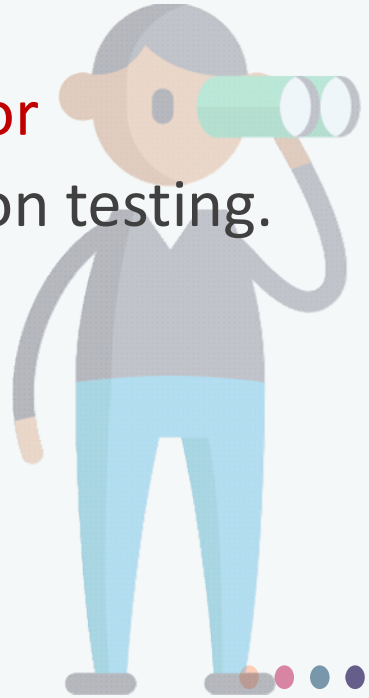


Exploratory Testing

- A hands-on approach in which testers are involved in minimum planning and maximum test execution.
- Test design & execution activities are performed in parallel, often without formal documentation.
- Informal tests are designed, executed, logged, and evaluated dynamically during test execution.
- The test results are used to learn more about the component or system, and to create tests for the areas that may need more testing.

Exploratory Testing

- Exploratory testing is sometimes conducted using session-based testing to structure the activity.
- In session-based testing, exploratory testing is conducted within a defined time-box, and the tester uses a test charter containing test objectives to guide the testing.
- Exploratory testing is most useful when there are few or inadequate specifications or significant time pressure on testing.



Checklist-based Testing

- Experienced tester uses a checklist to design, implement and execute tests based on the items or test conditions in the list.
- As part of analysis, testers create a new checklist or expand an existing checklist, but testers may also use an existing checklist without modification.
- Such checklists can be built based on:
  - experience of the tester
  - knowledge about what is important for the user
  - understanding of why and how software fails

Choosing Test Techniques

Choosing test techniques

- Formality. The use of techniques can vary in formality, depending on several factors:
  - Safety or regulatory industries
  - The maturity of the organisation
  - The life cycle model
  - Knowledge and skills of testers

Choosing test techniques

Factors influencing the decision about which technique to use:

- Type of component / system
- Component / system complexity
- Regulatory standards
- Customer or contractual requirements
- Risk levels and Risk types
- Test objectives
- Available documentation

Choosing test techniques

Factors influencing the decision about which technique to use (continued):

- Tester knowledge & skills
- Available tools
- Time & budget
- SDLC model
- Expected use of the software
- Previous experience with using the test techniques
- Expected types of defects

Choosing a test technique

- The internal factors that influence the decision about which technique to use are:
  - *Models used*
  - *Tester knowledge/experience*
  - *Likely defects*
  - *Test objective*
  - *Documentation*
  - *Life cycle model*

Choosing a test technique

- The external factors that influence the decision about which technique to use are:
  - *Risk*
  - *Customer/contractual requirements*
  - *Type of system*
  - *Regulatory requirements*
  - *Time and budget*